

AIrtist: Agentic Concept Development for Contemporary-Art Installations

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Abstract. Early-stage contemporary-art installation development often begins from an under-specified artist-authored problem rather than a resolved artifact. We introduce AIrtist, an agentic system that develops such briefs into five complete project hypotheses through persistent proposal lineages, domain-structured review, contestable intervention, and separate terminal ranking. Each incumbent is reviewed into an opportunity state that distinguishes what should be preserved from what should change; Focused and Structural challengers then compete with the unchanged incumbent rather than automatically replacing it. We evaluate AIrtist on 50 held-out commission-style briefs using exhaustive blind 5×5 candidate-pool comparisons, exact A/B presentation reversal, and a separately instantiated GPT-5.6 Sol holistic pairwise evaluator. Against a substantially matched shared-parent generic iterative control, AIrtist yields a centered complete-pool effect of +0.0836 (95% CI [+0.0340, +0.1304], Holm $p = .0015$). Across 250 lineages, 149 final states remain incumbent-identical; among 101 adopted replacements, challengers receive mean exact-parent preference 0.693 and a brief-balanced centered effect of +0.437. Terminal selection is less resolved: Hybrid’s +0.0533 Utility@1 advantage over Base is not statistically resolved. These results separate candidate-set construction, intervention quality, and attention allocation as distinct empirical problems. The evidence concerns textual proposals under model-mediated evaluation; it does not establish professional curatorial judgment, realized-artwork quality, or human co-creative benefit.

Keywords: contemporary art; installation art; concept development; creative agents; computational creativity; human-AI co-creativity; agentic AI.

1 Introduction

A consequential part of contemporary-art practice occurs before there is an artifact to fabricate, render, or exhibit. An artist may begin with a pressure rather than an image: a social relation that should become physically operative, a site whose existing use should not be erased, a material restriction, an institutional contradiction, or a conviction that several obvious artistic solutions would be wrong. At this stage, the artist may know what matters without yet knowing what the work should be. The problem is not how to execute a resolved concept, but how to develop alternative propositions for what the artwork could become.

Recent generative models make this stage newly accessible to computational support. They can produce plausible project descriptions, variations, references, images, and technical elaborations rapidly. Yet abundance of plausible completions does not by itself solve early concept development. In a contemporary-art installation, the same premise can lead to projects with radically different artistic operations: a concern may be represented symbolically, made causal through a material process, embedded in a spatial condition, delegated to an institutional procedure, or enacted through a spectator’s position. A proposal can become clearer, longer, more interactive, or more technically sophisticated while becoming artistically weaker. Several incompatible directions may also remain viable at once. The computational problem is therefore not simply to generate more material, but to construct useful alternatives, decide what in an existing proposal should survive intervention, and recognize which resulting possibilities deserve attention.

Installation practice makes these difficulties particularly visible because the identity of a work can depend on relations among material, bodies, space, duration, site, and institutional framing rather than on a detachable artifact alone [1, 2]. We use artistic operation for the relation or mechanism through which a proposal enacts its stakes: what the work actually does through its

material, spatial, temporal, spectator, or institutional organization. This does not define artistic quality, but it gives concept development a more discriminating computational object than topic, style, or visual appearance. A system working at this stage must reason not only about whether a proposal is coherent, but about which relation gives it identity and what kinds of change would preserve, sharpen, or destroy that relation.

Recent artistic practice already shows generative systems participating while the identity of a future artwork is still being articulated, including Kurant’s Errorism [3], Lund’s In the Middle of Nowhere [4], and Cai Guo-Qiang’s cAI [5, 6]. Section 3 examines these precedents in detail.

These practices establish the use case but remain artist-specific procedures embedded in particular oeuvres and projects. Adjacent computational research, meanwhile, provides sophisticated machinery for early ideation, including professional concept design and scientific hypothesis generation, iterative review, and multi-candidate development [7, 8, 9]. What remains largely unformalized is the task between them: taking a previously unseen, under-specified contemporary-art problem and systematically developing several complete proposals for what the artwork itself could become.

We introduce AItist, an agentic system for this task. An artist supplies a rough contemporary-art installation brief describing the pressure they want to investigate, conditions that must remain consequential, directions they want to avoid, and—where relevant—site, material, institutional, or production constraints. We use concept development for this domain-facing task; computationally, AItist represents it as search over persistent proposal states. The system treats the brief as a provisional problem specification rather than as a description of a desired artifact, develops five complete project hypotheses, allows existing states to survive critique, and finally ranks the resulting five-proposal set.

To our knowledge, AItist is the first evaluated computational system whose primary task is to develop an under-specified artist-authored contemporary-art installation brief into multiple complete hypotheses for what the artwork itself could become. The computational object is the artwork proposition itself: an artistic operation together with its material, spatial, spectator, field, and institutional realization.

A held-out example illustrates the distinction. Brief 053 asks for an installation about contested maintenance using canvas, galvanized mesh, and hand-mixed pigment, with every component repairable in public during opening hours. In the parent proposal, Echoing Gutter, a six-meter galvanized-mesh curtain spans a corridor. Pigmented water is released onto the mesh, and a visitor uses a broad brush to spread the wet streak back across the surface until the concentrated mark becomes less visible. The cycle makes damage and repair public, but it is largely reversible: each intervention returns the mesh toward its previous appearance, while canvas—although required by the brief—remains peripheral.

AItist’s selected Structural challenger, Layered Ledger, reorganizes the installation as mesh panels with pigment-receptive canvas behind them. A fresh pigment line is again released onto the mesh. Brushing now has two simultaneous effects: it redistributes the concentrated wet streak across the front while driving part of the pigment through the mesh openings toward the canvas, where it is absorbed and leaves a persistent ghost trace. Each cycle therefore returns the front toward visual uniformity while irreversibly adding to the record behind it. The artistic operation changes from damage → repair → reset to damage → repair → {front returns toward uniformity, history accumulated}. The intervention changes what maintenance physically does.

This example also exposes why generic iterative refinement is an incomplete model of the task. A critique may correctly identify a weakness while the proposed repair destroys the strongest relation in the parent. Conversely, a proposal may contain a valuable operation even when some part of its realization should be replaced. Additional inference should therefore be able to test a possible transformation without making transformation mandatory. Likewise, producing several proposals creates a second problem: a strong direction may already exist in the candidate set while a ranking policy fails to surface it. Concept development, intervention, and selection are consequently related but empirically separable problems.

AItist operationalizes this view through five persistent proposal lineages. Each incumbent first receives heterogeneous specialized review. The resulting evidence is compiled into an opportunity state that identifies what should survive, the dominant liability, a familiar field logic to escape, an unused material or spatial opportunity, a decisive transformation to attempt, and additive fixes that should not count as progress. Two challengers are then generated independently from the same frozen parent. A Focused reconstruction rebuilds the diagnosed weak part while preserving the incumbent property marked for protection; a Structural escape is additionally permitted to replace the dominant operation or realization when staying closer

would preserve the diagnosed problematic logic. Crucially, the incumbent remains eligible. Generating a challenger therefore does not require accepting it. After one state survives in each lineage, a separate terminal selector ranks the five resulting proposals.

The evaluated system is intentionally autonomous within this bounded stage: the artist provides the brief once and receives five ranked project hypotheses without intervening during the run. AIRTIST is therefore a concept-development backend rather than a complete conversational co-creative partner. Negotiation over diagnoses, criteria, branches, and longer-term artistic memory is left to future artist-facing systems.

The experimental design follows the same decomposition. RQ1 asks whether domain-structured development produces a stronger five-proposal set than raw generation, a generic self-refinement pass, and a substantially matched generic iterative process. The strongest control, M5, starts from the exact same five self-refined parent proposals as AIRTIST, maintains the same number of lineages and challengers, keeps the incumbent eligible, and shares challenger gating and lane comparison, while removing AIRTIST’s specialized review, opportunity representation, and differentiated intervention roles. RQ2 follows proposal identity through development and asks whether replacements adopted by the frozen policy are externally preferred to the exact parent states they displace. RQ3 holds the generated five-proposal set completely fixed and changes only the terminal ranking policy, isolating recognition from generation.

We evaluate the system on 50 held-out commission-style briefs. For RQ1, every AIRTIST proposal is blindly compared with every proposal in each of three control pools, yielding complete 5×5 cross-pool matrices rather than evaluation of only system-selected winners. Each logical comparison is repeated under exact A/B presentation reversal. The primary external evaluator is a separately instantiated GPT-5.6 Sol holistic pairwise judge that does not observe system identity, lineage history, internal scores, or the external candidate utilities later derived from its judgments. Statistical inference is performed over briefs rather than individual candidate comparisons.

The results distinguish the three stages rather than producing a uniform end-to-end success story. For candidate-set construction, AIRTIST significantly exceeds all three planned controls after Holm correction. Against the substantially matched M5 condition, the centered complete-pool effect is +0.0836 (95% CI [+0.0340, +0.1304], Holm $p = .0015$); secondary diagnostics indicate that the gain is not confined to one oracle-best candidate. For intervention, 149 of 250 final lineage states remain content-identical to their incumbents. Among the 101 replacements actually adopted, challenger preference against the exact displaced parent is 0.693, with a brief-balanced centered effect of +0.437 (95% CI [+0.266, +0.594]). Selection is less resolved. The Hybrid selector improves mean Utility@1 by +0.0533 over Base on identical candidate pools, but the confidence interval includes zero; it identifies an externally highest-utility candidate first on only 26% of briefs. The experiment therefore provides strong evidence for improved candidate-set construction and adopted intervention quality while identifying terminal recognition as a distinct unresolved problem.

These results suggest a broader diagnostic view of creative agents. A weak surfaced proposal can arise because the system failed to construct strong alternatives, because an intervention damaged an existing state, or because ranking hid useful material that had already been generated. Evaluating only a final top-ranked output collapses these failure modes. Contemporary-art concept development makes the distinction especially visible because a change in material causality, spatial organization, spectator position, or institutional procedure can alter the identity of the proposal rather than simply improve its presentation. Generative assistance can induce fixation [10] and, in other settings, improve individual outputs while narrowing collective diversity [11], reinforcing the need to distinguish candidate quality from the mere existence of multiple branches.

The paper makes four main contributions:

- 1. Contemporary-art concept development as an AI task.** We formulate and evaluate early-stage installation concept development from under-specified artist-authored briefs, where the computational object is a set of alternative project hypotheses rather than the realization of an already resolved artwork.

- 2. A contestable agentic development architecture.** AIRTIST maintains persistent proposal lineages, converts heterogeneous review into an explicit opportunity state, generates Focused and Structural interventions from the same incumbent, and allows the incumbent to survive. The architecture represents what should remain invariant separately from what may need to change.

3. A multi-level evaluation methodology for open-ended creative agents. We separately evaluate the quality of the sampled possibility set, adopted interventions against their exact parents, and terminal ranking over identical generated candidate sets, including a substantially matched shared-parent iterative control.

4. Empirical evidence about where agentic value and error arise. Domain-structured development improves measured candidate-set quality over matched generic iteration; adopted replacements are preferred to the exact states they replace on average; and terminal selection remains an independent recognition problem even when stronger measured alternatives already exist in the generated set.

The claims remain bounded by the evaluation setting. AIArtist does not establish autonomous artistic agency, professional curatorial judgment, or improved human creativity. The evaluated objects are textual project hypotheses, the principal quality signal is model-mediated, and no artist negotiates with the system inside the frozen experimental pipeline. The present study instead asks whether early contemporary-art concept development can be made into an explicit computational task and whether the proposed agentic development policy produces measurable improvements under a disclosed evaluation procedure.

Paper organization. Section 2 develops the contemporary-art formulation underlying the task, including artistic operation and the bounded working quality model. Section 3 positions AIArtist relative to contemporary artistic practice, AI-assisted ideation, and agentic refinement. Section 4 describes the AIArtist architecture, and Section 5 specifies the benchmark, controls, external evaluation, and inference procedure. Sections 6 and 7 report the main results and mechanism/failure analysis. Section 8 discusses implications for creative-agent design; Section 9 develops the intended artist-facing use and future negotiated co-creativity; Section 10 states limitations and threats to validity; and Section 11 concludes.

2 Representing Contemporary-Art Installation Ideation as a Computational Problem

AIArtist does not generate a physical installation. It generates a proposal for a future installation: a textual account of what would be installed, how its material and spatial organization would behave, how spectators would encounter it, what institutional conditions it would depend on, and why those choices belong to the artistic proposition rather than merely to its fabrication. The computational object in this paper is therefore not the realized artwork but a sufficiently developed textual dossier representing an installation hypothesis. We use *dossier* for a proposal developed enough to support further concept development and comparative judgment, not for a fabrication-ready specification. Further artistic, curatorial, engineering, conservation, and site-specific decisions remain expected.

The evaluated pipeline is autonomous after brief submission, but its intended artist-facing role is co-creative rather than substitutive. It develops candidate directions that an artist may inspect, reject, combine, reinterpret, or develop further. The method does not assume that the model captures the artist's tacit judgment, broader practice, or all physical, material, and institutional contingencies of realization. A realized installation may therefore depart substantially from its dossier through scale, fabrication, material accident, maintenance, institutional negotiation, audience behavior, or later artistic decisions. Such departures are compatible with the intended workflow rather than failures to reproduce a fixed specification.

A separate difficulty concerns representation within the textual task. A language model can readily generate a title, describe a room and a set of objects, propose a material palette, and produce persuasive curatorial prose. None of these, separately or together, guarantees that the text specifies a coherent installation. Two dossiers may address the same topic and use almost the same materials while proposing fundamentally different works because bodies move differently through them, matter changes differently over time, access is withheld under different conditions, or the institution plays a different role. Conversely, works that look unrelated at the level of object vocabulary may depend on closely related temporal, material, spatial, perceptual, spectator, or institutional dependencies. For installation ideation, representing the components is therefore not enough: a small change in their relations can change what the proposed work does.

The study therefore makes a limited representational assumption: a textual dossier can describe a proposed installation in enough detail to support meaningful computational reasoning about the relations through which the work is intended to operate. "Enough detail" here has a practical meaning. The dossier should state what exists, what changes, what depends on what,

what a spectator can perceive or do, which material properties matter, and which production or institutional conditions are necessary. It need not—and in practice cannot—determine every property of the eventual artwork.

For proposal development, we distinguish three analytical levels:

- **Artistic stakes:** What tension, contradiction, experience, or question makes the proposed work worth pursuing?
- **Artistic operation:** Which specified dependency makes that tension occur in the proposed work rather than merely being named by it?
- **Realization:** What concrete materials, spatial arrangement, timing, technical system, and institutional conditions instantiate that operation here?

These are not claimed to be ontological components present in every artwork. They form a computational scaffold for proposal development: three aspects that the system must be able to distinguish if it is to alter one part of a proposal without inadvertently erasing another.

The section develops this representation in two stages. First, installation theory and established artistic practice motivate why space, duration, material behavior, embodied spectatorship, site, and institutional context may be constitutive rather than incidental. Second, we translate those concerns into an explicit, bounded computational surrogate of proposal quality for one task: developing contemporary-art installation proposals under museum-commission conditions. The theoretical literature motivates the kinds of relation that the representation should be able to express; it does not directly determine the computational quality model introduced later or how AIArtist operationalizes it.

2.1 Installation as a Situated Configuration

A useful starting point is that installation is rarely exhausted by a self-contained object. Bishop foregrounds the spectator's bodily presence within the work; Petersen describes installation as activating physical space and context, extending the work in time, and emphasizing embodied perception; and Kwon traces a transformation of site from fixed physical location toward institutional, social, and discursive conditions [1, 12, 2]. O'Doherty's analysis of the white cube makes a complementary point: gallery space actively frames the conditions under which something is encountered as art rather than serving as a transparent container [13]. Rebentisch treats site-specific installation as engaging the interrelation of literal and social sites [14]. These positions are not interchangeable, and AIArtist does not combine them into a single definition of installation. Morris's account of sculpture and Burnham's systems aesthetics make the move away from the isolated object especially explicit: placement, surrounding space, the viewer's changing position, and relations among people and environmental components can become constitutive conditions of the work [15, 16]. For the present computational problem, this literature supports a narrower representational premise: an installation may be constituted not only by the intrinsic properties of objects but by dependencies among material, space, duration, spectatorship, and institutional context.

We do not adopt these theoretical positions as a ready-made computational definition. Their relevance is practical. They warn against representing an installation as a list of objects plus a theme. If changing a route, a temporal condition, a spectator position, a material behavior, or an institutional rule can change what the work does, those elements must be available to the computational system as first-class parts of the proposal.

Table 1 makes this representational consequence concrete across ten heterogeneous works. It does not propose a taxonomy of installation or attribute AIArtist's terminology to the artists. The physical-encounter column summarizes documented features of each work in enough detail to make the relevant relation intelligible without prior familiarity with the work; the artistic-operation column gives our analytical reading of one constitutive dependency relevant to the present representation rather than an exhaustive account of the work; and the final column identifies the realization conditions on which that dependency relies.

Table 1: Artistic operations and their realization across heterogeneous installations.

Work	What a visitor physically encounters	Artistic operation	Realization conditions
James Turrell, <i>Frontal Passage</i> (1994) [view]	A visitor passes through a darkened entryway into a dark chamber. Concealed light sources produce a radiant, sharply bounded field of red light across the space. The source itself is not visible; the apparent boundary is light, yet it seems to end abruptly in space as if it had density or formed a physical plane. [17]	<i>Situated viewer ↔ controlled light field</i> : under these viewing conditions, immaterial light is perceived as a bounded, apparently physical presence.	Darkened entry and chamber; concealed light sources; a sharply controlled red-light field; a viewer positioned within the constructed viewing environment.
Janet Cardiff, <i>The Forty Part Motet</i> (2001) [view]	Forty separately recorded choir voices are played simultaneously through forty individual loudspeakers arranged in eight groups of five around a large oval. Moving close to one speaker isolates an individual singer; standing between speakers produces combinations of nearby voices; from the center, the listener can hear the forty voices together as a choir. [18]	<i>Listener position ↔ spatially separated voices</i> : movement through the speaker field changes the audible relation between individual singers, local combinations of voices, and the choir as a whole.	Forty separately recorded vocal tracks; one loud-speaker per voice; spatial separation of the speakers; a navigable listening space in which position changes the heard mix.
Mel Bochner, <i>Measurement Room</i> (1969) [view]	The actual gallery room is measured, and strips of black tape and numerical notation mark the dimensions of the same walls, doorways, windows, and passageways that the visitor is standing among. The visitor therefore occupies a room whose physical dimensions are displayed directly on the architecture being measured. [19]	<i>Actual architecture ↔ numerical description of that same architecture</i> : the room is encountered simultaneously as physical space and as a measured representation of itself.	The actual room; measurements derived from that room; notation applied directly to the architectural elements being measured; the viewer inhabiting the measured space.
Doris Salcedo, <i>Shibboleth</i> (2007) [view]	A long, irregular fissure opens through the floor along Tate Modern's Turbine Hall. It varies in width and depth, and its concrete sides reveal steel mesh within the rupture. Rather than appearing as a separate object placed in the hall, the fissure occupies the floor across which visitors themselves move. [20]	<i>Museum circulation ↔ rupture in the floor being traversed</i> : a surface ordinarily encountered as continuous is physically interrupted, making division part of the visitor's bodily passage through the hall.	The actual Turbine Hall floor; an extended fissure within that circulation surface; sufficient width, depth, and material presence for the rupture to be physically encountered; visitor movement through the same space.
Hans Haacke, <i>Condensation Cube</i> (1963–67) [view]	A transparent acrylic or methacrylate cube contains water in a closed container. Light warms the interior; water evaporates and condenses on the inner walls as droplets that grow, merge, and eventually run down the transparent surfaces. The patterns continually change with light and temperature in the surrounding exhibition space, and even nearby bodies can slightly alter those conditions. [21, 22]	<i>Contained water system ↔ local exhibition environment</i> : changing light and temperature conditions continually produce different condensation states and therefore different visible states of the work.	Transparent enclosure; contained water; a temperature relation between interior and surrounding space; light and local environmental conditions; duration for evaporation, condensation, droplet growth, and movement to occur.
Bruce Nauman, <i>Corridor Installation (Nick Wilder Installation)</i> (1970) [view]	Six narrow corridors of different widths are constructed within the gallery; three can be entered and three cannot. In one cramped passage, stacked video monitors are visible at its far end. As a visitor moves through the installation, closed-circuit cameras produce views that differ conspicuously from the body's direct experience of its position and movement. [23]	<i>Constrained bodily movement ↔ displaced surveillance view of that movement</i> : the visitor occupies and advances through one spatial position while simultaneously encountering a mediated image of that position from elsewhere.	Narrow, partially accessible corridors; constrained routes; closed-circuit camera coverage from different viewpoints; monitors positioned within the installation; a live relation between bodily movement and displayed image.
Felix Gonzalez-Torres, "Untitled" (Portrait of Ross in L.A.) (1991) [view]	Individually wrapped, variously colored candies are arranged in the exhibition space. The work specifies an ideal weight of 175 lb and an endless supply, while its dimensions and configuration may vary between manifestations. Visitors must be allowed to choose whether to take candies, so the quantity physically present can diminish; during an exhibition, authorized decisions determine whether and how the candies are replenished and maintained. [24]	<i>Visitor removal ↔ changing material quantity ↔ institutional maintenance</i> : spectators can physically diminish the manifested work, while its subsequent material state also depends on decisions about replenishment.	Individually removable candies; permission for visitors to take them; a material quantity capable of changing through removal; an endless supply; authorized decisions about replenishment and maintenance.
Ann Hamilton, <i>the event of a thread</i> (2012) [view]	A field of swings fills Park Avenue Armory's Drill Hall. Through ropes and pulleys, the swings are connected to the same enormous white cloth suspended high across the hall. As different visitors swing, their separate movements combine through the shared rigging, causing the single cloth to rise, fall, ripple, and billow. Readings, sound, pigeons, writing, and other live elements also occupy the larger installation. [25]	<i>Individual bodily movements ↔ shared mechanical network ↔ common material surface</i> : actions distributed across many swings combine physically and become visible in the changing movement of one cloth.	Visitor-accessible swings; a shared rope-and-pulley system; mechanical coupling of the swings to the same suspended cloth; sufficient suspension and travel for riders' movements to propagate into the fabric.

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Table 1 continued

Work	What a visitor physically encounters	Artistic operation	Realization conditions
Michael Asher, <i>Kunsthalle Bern, 1992</i> (1992) [view]	Visitors entering the Kunsthalle encounter radiators removed from the building's exhibition rooms and regrouped in the entrance gallery. The radiators remain functional: multiple steel pipes run from them through the building to their original heating connections. The entrance therefore contains a concentrated field of working radiators, while the exhibition rooms from which they were removed are left without their usual heating elements. [26, 27]	<i>Institutional heating infrastructure</i> ↔ <i>spatial redistribution of heat</i> : re-locating the museum's functioning radiators makes normally background support infrastructure physically perceptible while changing the thermal conditions through which different parts of the building are encountered.	The Kunsthalle's actual radiators; their removal from the exhibition rooms; continued connection to the working heating system; concentration of the radiators in the entrance gallery; the resulting redistribution of heat across the building.
Precious Okoyomon, <i>Among the Flowers I Learned to Love: A Garden of Decreation</i> (2026) [view]	MoMA PS1's outdoor courtyard is transformed into a living garden or forest containing grasses, flowers, weeds, sedges, vines, pollinators, sculpture, and sound. Native and invasive, annual and perennial, cultivated and wild species grow together. A giant teddy-bear sculpture at the center can be entered and climbed through. Installed from 2026 through 2028, the garden continues growing and changes with the seasons rather than remaining in one fixed material state. [28]	<i>Living plant assemblage</i> ↔ <i>seasonal and environmental change</i> : growth and changing conditions continually alter the material state of the installed landscape and therefore what occupies the courtyard over time.	Living mixed plant species; outdoor exposure to seasonal and environmental change; the site-specific courtyard landscape; sustained duration allowing growth and seasonal transformation to alter the installed state.

The examples differ radically in medium, scale, temporality, and spectator agency, but they make the same representational point visible. A viewer may only look or listen, may move through a constrained spatial relation, may remove material, or may set a shared mechanism in motion; in other cases, the relevant change is produced by climate, institutional infrastructure, or the growth of living matter. Neither participation, technology, material transformation, nor institutional procedure therefore defines artistic operation. What matters computationally is narrower: when changing a relation among material, space, duration, spectator, or institution would change what the installation does, that relation must be represented explicitly enough to be preserved, challenged, or replaced during proposal development.

2.2 Artistic Operation

For AIRTIST, an artistic operation is a specified dependency through which a proposal enacts its artistic stakes rather than merely naming them. In plainer language, it answers: What actually happens because this installation has been arranged in this particular way? What changes, for whom or for what, and what follows from that change? The participating terms of the dependency should therefore be identifiable.

The term is a computational abstraction used by AIRTIST, not an established art-historical category. It gives the system a level of description between stakes—too abstract by themselves to guide revision—and realization—too specific by itself to tell us what should survive when particular materials or mechanisms change. A stake such as loss or displacement is not itself an operation, just as candy, a camera, or a loudspeaker is only part of a realization. The operation lies in the relation through which such elements acquire a specific role in what the work does. The same stakes can be enacted through different operations, and similar operations can support different stakes.

A useful way to clarify this distinction is to ask which changes in realization would preserve the relevant dependency and which would remove it. In Cardiff's *The Forty Part Motet*, the particular room, loudspeaker models, and precise spacing could vary within limits so long as each recorded voice remained spatially individuated and a listener's movement continued to change which singers were locally foregrounded and how the choir recombined. Conversely, the same forty recordings and the same forty loudspeakers could remain while every loudspeaker reproduced the same premixed choir. Much of the sound material and even the visible arrangement would be retained, but movement would no longer disclose different individual voices or local groups; the spatial dependency identified here would be gone [18]. Haacke's *Condensation Cube* provides a material counterpart. A transparent cube could still show changing droplets if a concealed programmed system periodically produced the visible droplet pattern rather than allowing evaporation and condensation to generate it in response to local environmental conditions. The visible motif and temporal change could remain even though the environmental dependency had been removed [21, 22]. These counterfactuals do not test artwork identity; they separate variable realization details from the dependency that defines the operation.

2.2.1 Operation is not mechanism

Generative models can easily turn a topic into a device—surveillance into a camera, memory into an archive, climate into sensors or live data—and then retrofit curatorial prose that makes the device sound conceptually necessary. AIRTIST therefore distinguishes the artistic operation from the mechanism that realizes it. A mechanism may be materially or technically conspicuous without specifying what the work does; its relevance depends on the relation it establishes or sustains.

Nauman's *Corridor Installation* (*Nick Wilder Installation*) makes the distinction concrete. Its cameras and monitors are realization components; in the present reading, their artistic role lies in coupling constrained bodily movement to a live view of that movement from a spatially displaced position. The same camera-monitor system could function as ordinary surveillance without producing that relation, while different imaging hardware could in principle preserve it. Changing a mechanism therefore need not change the operation, provided the relevant dependency survives [23]. A concrete AIRTIST case in which an artistic relation is preserved while its mechanism must change is developed through the Delayed Projection example in Sections 4.3–4.4.

2.2.2 Operation is not participation

Participation is likewise neither a prerequisite for nor an adequate description of an artistic operation. In Turrell's *Frontal Passage*, the spectator need only enter, look, and move within a precisely constructed viewing relation. In Cardiff's *The Forty Part Motet*, listener position changes which separately spatialized voices are locally foregrounded without requiring the visitor to contribute content or control the system. Hamilton's *the event of a thread* is participatory in a stronger causal sense: visitors' swinging is transmitted through shared rigging into the movement of a common suspended cloth. For the present representation, these are not positions on a scale from passive-and-weak to interactive-and-strong; they are different dependencies assigning different roles to spectator position and action [17, 18, 25].

This distinction matters computationally because generative systems frequently answer criticism about “engagement” by adding buttons, sensors, choices, visitor submissions, or responsive media. AIRTIST gives no automatic bonus for these additions. Its spectator-specificity criterion asks instead how the spectator is positioned in the work's logic, what can or cannot happen from that position, and what changes because the spectator is there.

This refusal of a simple passive/active hierarchy has clear precedents. Ranciere explicitly re-examines the inherited equivalence between looking and passivity, while Bishop calls the binary of active and passive spectatorship unhelpful and rejects participation as an automatic aesthetic or political virtue [29, 30]. These positions do not define AIRTIST's spectator metric, but they support its refusal to equate observable activity with quality.

2.2.3 Why AIRTIST asks for one dominant operation

AIRTIST deliberately imposes a stronger representational constraint here than a purely descriptive theory of installation would require. The structured dossier contains a single `artistic_operation` field, and the generation contract asks each candidate to identify one dominant operation. A supporting operation is not a second dossier field: it means a secondary dependency that is causally necessary to the dominant operation and serves it rather than establishing a competing artistic logic. The default is therefore one dominant operation and normally no more than one supporting operation. This is part of AIRTIST's search representation rather than explanatory vocabulary added after the fact.

The constraint is not an ontological claim that a realized artwork possesses one objectively true operation. An installation may sustain several simultaneous, historically unstable, or mutually irreducible relations. The computational requirement is narrower: during revision, AIRTIST needs a current dependency against which a proposed change can be understood as preserving, strengthening, or replacing the artistic logic under development. Without such a conservation target, a dossier can become more elaborate while the relation that made the proposal worth developing drifts or disappears.

The limit is a default economy guardrail rather than a ban on complexity. Additional dependencies are acceptable when they are necessary to the governing operation, when removing them would cause meaningful artistic loss, or when the realization makes them unavoidable; complexity itself receives no bonus. Thus “one dominant operation” does not mean one object, one effect, or one interpretation. It means that additional mechanisms or relations should remain subordinate to the dependency the proposal is organized around rather than accumulating

as parallel features.

2.3 A Bounded Working Model of Proposal Quality

The argument so far concerns what an installation proposal needs to represent. It does not yet tell Airtist how to search for a stronger proposal. Installation theory motivates attention to situated space, material behavior, duration, spectatorship, and context. It does not imply Airtist’s ten computational dimensions, their decomposition, their use in ranking, or the precise search pressures encoded by the system. Those are methodological choices. Airtist therefore operates under an explicit bounded surrogate of installation-proposal quality. It is a computational and curatorial hypothesis about properties that make the class of museum-installation proposals studied here worth developing.

Computational-creativity research provides a methodological precedent for making evaluative commitments explicit. Ritchie proposes observable empirical criteria for judging the behavior of creative programs, while Jordanous’s SPECS procedure more directly begins by stating what creativity means for the particular system being evaluated and then deriving tests from that account [31, 32]. Airtist follows the same methodological principle without claiming that its chosen criteria exhaust artistic quality.

Airtist operationalizes this surrogate through ten equally weighted dimensions. In the evaluated configuration, each dimension is constrained to the interval $[0, 1]$, and their arithmetic mean forms the uniform aggregate used in quality-bearing comparison. This numeric structure is a search-and-ranking device, not a claim that artistic quality is naturally ten-dimensional or that exact semantic thresholds exist at 0 and 1. The dimensions function as bounded internal pressure tests and scoring constructs; proposal generators are explicitly instructed not to self-score or recite them as curatorial rhetoric.

Table 2 gives qualitative low- and high-end anchors for all ten dimensions. The anchors summarize the direction of the implemented rubrics rather than defining deterministic numerical cutoffs.

Table 2: Qualitative low- and high-end anchors for the ten proposal-quality dimensions.

Dimension	Low end	High end
Raw-prompt fidelity	Answers a safer adjacent theme or loses the brief’s concrete pressure, unusual demand, or explicit constraints.	Preserves the brief-specific pressure and constraints closely enough that the proposal remains a response to this problem rather than a generic version of its topic.
Core-tension enactment	States or illustrates the central tension in thesis-level prose or metaphor, while the installed form does little to make that tension happen.	Makes the central tension operative through what the installation materially, spatially, perceptually, temporally, socially, or procedurally does.
Artistic-operation necessity	Depends mainly on explanation, a generic mechanism, a decorative effect, or a familiar metaphor whose installed form is readily interchangeable.	Depends on a specific operation, relation, or form whose removal or substitution would cause meaningful artistic loss.
Material necessity	Uses material or realization mainly as a replaceable carrier; a generic alternative would leave the operation substantially unchanged.	Depends on specific properties, behavior, process, history, or institutional relations of the realization; variation is compatible with high necessity when the constitutive property or rule is preserved.
Spatial encounter	Could be translated into a text, image, object, or generic display with little loss because spatial arrangement and duration do not materially alter the encounter.	Requires a concrete spatial-temporal relation in which position, access, distance, sequence, duration, scale, or another situated condition changes what the work makes available.

Continued on next page

Table 2 continued

Dimension	Low end	High end
Visual encounter	Is perceptually generic, inert, or diagrammatic, or gains visual force mainly from detachable spectacle that is not necessary to the work.	Produces a memorable perceptual and spatial situation whose visual or material force follows from the installation's actual operation rather than replacing it.
Spectator specificity	Leaves the spectator as a generic viewer or participant whose position, attention, action, duration, access, or knowledge makes little difference.	Places the spectator in a precise and consequential relation in which position, attention, action or non-action, access, duration, visibility, knowledge, or relation affects the encounter.
Field distinctiveness	Repeats a familiar artistic format or trope with a new topic, material, style, or atmosphere while leaving its underlying operating logic largely unchanged.	Makes a specific structural transformation of a recognizable artistic format or trope; absolute novelty is not required.
Public-reception resilience	The most available public reading reduces the work to a familiar generic signal or format and largely exhausts its more specific artistic tension.	Simplified public readings remain possible, but they do not exhaust the installed relation or eliminate the more specific artistic pressure it produces.
Production/institutional viability	The central operation depends on major unacknowledged production or institutional requirements that make the proposal implausible under the stated commission conditions.	The central dependency can plausibly be developed within the stated institutional and production envelope, with ordinary later engineering or readiness work remaining addressable.

For recurring prompt-level guidance, AItist also uses a more compact five-principle quality contract covering brief-specific stakes, artistic-operation and form-concept necessity, visual-spatial identity, field-aware distinctiveness, and production/institutional viability. We use this coarser form as an engineering abstraction so shared generation and review prompts can carry common quality guidance without reproducing the full ten-dimensional rubric. It is not a second scoring model: AItist computes no five-dimensional aggregate, and the five principles are not a formal one-to-one partition of the ten dimensions.

The remainder of this subsection explains the ten dimensions and the distinctions among them.

2.3.1 Raw-prompt fidelity and core-tension enactment

These dimensions address a common language-model failure: producing a fluent work near the topic while escaping the actual problem. Raw-prompt fidelity asks whether the proposal preserves the concrete pressure, constraints, stakes, and unusual demands that make the brief specific. It is deliberately literal where the brief is literal: a proposal that uses cameras when cameras are prohibited should not recover merely because the resulting installation sounds sophisticated.

Core-tension enactment asks a different question. A proposal can satisfy every keyword while still avoiding the reason those constraints were introduced. A brief about unequal waiting, for example, can mention delay, clocks, and inequality while constructing little more than a visual metaphor for time. Enactment asks whether inequality in waiting changes the proposed situation—who waits, for how long, under what fixed condition, and what becomes perceptible because those durations differ. The two scores can therefore diverge: fidelity asks whether the proposal remains inside the problem the artist gave us; enactment asks whether that problem becomes structurally present in what the installation does.

2.3.2 Artistic-operation necessity and material necessity

Artistic-operation necessity asks whether the proposal depends on a specific organizing dependency rather than on explanatory prose, a generic mechanism, a decorative effect, or a familiar

metaphor. Put literally: if the curatorial paragraph disappeared, what would the installation itself still do that carried the proposition? At the high end, removing or substituting the dominant operation would cause meaningful artistic loss rather than merely alter presentation.

Material necessity asks whether a chosen material or realization system contributes through a consequential property, behavior, process, history, or institutional relation rather than functioning as an interchangeable visual sign. This is not literal medium essentialism. A variable, delegated, or site-adaptive realization can remain materially necessary when the specific property or dependency on which the work relies survives the change. Morris's account of sculpture provides an earlier historical motivation for judging physical decisions relationally: he shifts attention from isolated internal properties of the object toward relations involving placement, surrounding space, light, and the viewer [15]. This motivates the relational question but does not itself derive AIArtist's material-necessity score.

2.3.3 Spatial encounter, visual encounter, and spectator specificity

These three dimensions deliberately separate aspects that generative prose often bundles together as "immersive," "experiential," or "engaging." Spatial encounter asks whether the proposal establishes a necessary situation in space and time. Route and movement can matter, but they are not required: the relevant organization may instead depend on position, access, threshold, distance, scale, orientation, visibility, duration, an object or image encountered from a particular relation, a material process, or an atmosphere that unfolds in place. The question is not simply where a visitor walks, but what spatial-temporal arrangement must hold for the proposed encounter to occur.

Visual encounter asks whether the proposal produces a specific and memorable perceptual situation before explanatory prose is supplied. It is not an independent reward for beauty, spectacle, or photogenicity. The implemented rubric does, however, treat documentation or press-image potential as one bounded consideration: a work may be visually strong while resisting complete photographic translation, provided that resistance follows from a specific and necessary encounter rather than from visual indeterminacy. Turrell is a useful limiting case because the apparent solidity of red light is visually forceful while remaining dependent on controlled light, darkness, chamber, and viewing position rather than on accumulated spectacle [17].

Spectator specificity asks how the spectator is positioned in the logic of the work: through position, attention, movement, action or non-action, access, duration, visibility, knowledge, or consequence. Specificity therefore does not require participation. As Table 1 shows, the spectator may occupy a controlled viewing or listening position, move through a spatial relation, remove material, or causally alter a shared mechanism. What AIArtist rejects is the generic statement that "the viewer interacts" or "the viewer reflects" without specifying what that relation is materially, spatially, temporally, or institutionally contingent upon.

2.3.4 Field distinctiveness and public-reception resilience

Field distinctiveness is relational rather than a demand for novelty *ex nihilo*. Bourdieu's account of cultural production treats artistic and literary positions as intelligible differentially, through their relations to other positions and position-takings in a field [33]. Krauss's critique of avant-garde originality makes a different but complementary point: claims to originality can emerge against structures of repetition and recurrence rather than from a literal ground zero [34]. Groys makes the relational character of newness still more explicit by treating innovation through comparison with an existing cultural archive and through revaluation of what is already culturally recognized [35]. Bourriaud describes a historically more specific contemporary-art tendency in which artists work from pre-existing forms and produce singularity by reprogramming what is already available rather than beginning from a blank slate [36]. These accounts differ substantially, and none derives AIArtist's particular score. Together they support only the narrower premise needed here: artistic distinctiveness need not require an unprecedented object, medium, or formal vocabulary; it can arise through a consequential transformation of an already available artistic language, format, relation, or convention.

AIArtist therefore asks whether a proposal merely re-skins a recognizable artistic format or changes the relations through which that format operates. Gonzalez-Torres provides a concrete illustration within an established artistic language. Institutional accounts explicitly situate his candy works in relation to Minimalism, while emphasizing how visitor removal, changing quantity, and replaceability transform the assumptions of a stable, autonomous Minimalist object [37, 24]. The point is not that such transformation defines distinctiveness universally, but

that difference can be located in what an inherited artistic form is made to do rather than in whether its component objects or materials have never appeared in art before.

There is an important epistemic limit to this criterion. AArtist does not possess an authoritative map of contemporary art. Its field-context stage uses model-mediated recall of works, artists, formats, criticism, and neighboring contexts, not provenance-bearing bibliographic retrieval. Field evidence is therefore a working hypothesis about nearby artistic lines, useful for directing search but insufficient to establish an art-historical claim that a proposal is objectively unprecedented or distinct. In this paper, field distinctiveness means field-aware differentiation under the system's recalled context, not measured global novelty in contemporary art.

Public-reception resilience is an AArtist-specific proposal-level diagnostic rather than a construct derived from a general theory of audience reception. It asks whether a readily available generic framing captures almost everything the proposal actually does, or whether the installed relation retains consequential specificity that such a framing leaves unexplained. The criterion neither predicts how real audiences will interpret the work nor rewards opacity, difficulty, or resistance to popular reception for their own sake.

The distinction is visible in familiar surface descriptions. Gonzalez-Torres's work can be reduced to "a pile of candy," yet that phrase leaves out the relation among removable material, changing quantity, spectator choice, and institutional maintenance [24]. Hamilton's *the event of a thread* can be reduced to "a room of swings," yet that description omits the shared mechanical coupling through which distributed riders collectively move a common cloth [25]. These are illustrations of the diagnostic, not measurements of actual audience response. Eco's account of openness is relevant only as a boundary condition: interpretive plurality can coexist with work-specific structure, so resilience should not be confused with enforcing one correct interpretation [38].

2.3.5 Production and institutional viability

AArtist evaluates production/institutional viability because its computational object is a commission proposal for a future work. In the evaluated configuration, the criterion is judged relative to the stated museum-production context, including the approximate USD 150,000 project envelope described in Sections 4 and 5. This should not be confused with the proposition that feasible art is better art. Deliberate impossibility, waste, fragility, institutional conflict, or excessive labor can be artistically meaningful; they can nevertheless lower this particular score when the proposed realization is evidently impossible, radically outside the commission envelope, or dependent on an unacknowledged production contradiction. Ordinary engineering work that remains plausibly addressable is not itself a failure.

Viability therefore concerns the whole realization dependency rather than fabrication cost alone. If the artistic operation requires a maintenance rule, access to institutional infrastructure, permission, sustained labor, or a living system, the dossier should identify that dependency and make plausible how it can be supported under the stated commission conditions. Artistically intentional difficulty can remain part of the proposition; what lowers this score is an unacknowledged contradiction between the intended operation and the conditions required to realize or sustain it.

2.3.6 What the model deliberately does not reward

Beyond its positive criteria, the implemented quality contract explicitly withholds automatic credit for properties that language models can add cheaply. In particular, the prompts do not treat the following as independent evidence of a stronger proposal:

- surface novelty, a new material, or a different topic without a structural transformation of the artistic format;
- participation, audience engagement, or interactivity merely because the spectator is made active;
- technology, interface layers, sensory cues, or technical sophistication without structural necessity;
- complexity, detail, polish, completeness, or explicitness in themselves;
- visual spectacle or photogenicity detached from the work's visual-spatial logic;
- improved legibility or usability when it leaves the artistic operation unchanged or smooths away a productive unresolved relation;

- difficulty, maintenance burden, or institutional demand merely because it is difficult;
- feasibility or readiness improvements that weaken, replace, or fail to strengthen the artistic proposition.

This is not a prohibition on any of these properties. They can contribute when they are necessary to the dominant operation, brief-specific pressure, spectator relation, or visual-spatial identity. Documentation and press-image potential, for example, are a bounded consideration within visual encounter, but photogenic spectacle receives no independent bonus and a work can remain visually strong while resisting complete photographic translation. Likewise, the refinement prompts repeatedly reject fluency, completeness, feasibility detail, or clearer explanation as improvement in themselves and warn against answering critique by accumulating interfaces, signage, sensory effects, participation mechanics, or administrative/readiness layers. The point is to resist a generic refinement direction in which a dossier becomes more polished and feature-rich while the dependency carrying its artistic proposition becomes less necessary.

2.3.7 Ethical and public-context diagnostics

Ethical and public-context concerns are handled separately from the ten-dimensional quality vector. In the evaluated configuration, a dedicated `EthicsPublicContextReviewer` is assigned none of the ten scored dimensions and is explicitly diagnostic-only: its findings do not alter dimension scores, aggregate score, ordering, or winner selection. They are retained for final reporting, readiness assessment, known risks, and post-selection development. The reviewer examines issues including coercion and consent, extractive participation, vulnerable groups, labor, ecology, surveillance and data, contested histories, grief, public memory, testimony, access, and power relations.

This separation should not be read as a claim that ethical responsibility is irrelevant to artistic practice, nor as an attempt to convert ethical virtue into an eleventh aesthetic score. Bishop argues directly that social and artistic judgments do not easily merge and appear to require different criteria, while Kester’s dialogical aesthetics offers a contrasting evaluative locus in the character of the exchange itself [30, 39]. `AIrtist` adopts neither account as a scoring theory; their relevance here is narrower. They support keeping responsibility diagnostics visible without pretending that they are automatically commensurable with the ten-dimensional artistic-quality vector.

Table 3 closes the quality-model discussion by collecting the epistemic boundaries most likely to be confused with direct theoretical derivation or empirical validation. It is deliberately selective rather than a second inventory of all ten dimensions.

Table 3: Selected theoretical motivations and AItlist-specific computational commitments.

AItlist construct	Closest theoretical/practical motivation	Boundary inside AItlist
Situated proposal representation	Bishop; Petersen; Kwon; O’Doherty; Rebentisch; Morris	Installation and site discourse motivate representing embodied spectatorship, space, duration, site, and institutional framing as potentially constitutive. The cited authors make different claims; AItlist uses them as convergent motivation rather than as one unified theory of installation.
Artistic operation	Burnham’s relational systems; Morris’s expanded situation; installation-practice discourse	AItlist-specific abstraction for an explicit dependency through which a proposal does something rather than merely naming a topic. No cited theorist is claimed to supply the <code>artistic_operation</code> field or its one-dominant-operation constraint.
Ten-dimensional quality surrogate	Ritchie; Jordanous; task-specific curatorial criteria	Implemented $[0, 1]$ scoring vector with equal weights in the evaluated configuration. Computational-creativity work motivates explicit evaluative commitments; it does not validate this particular ten-dimensional model.
Field distinctiveness	Bourdieu; Krauss; Groys; Bourriaud	These sources support different aspects of a relational account of artistic difference: positions are field-relative, originality need not imply a ground zero, and innovation can work by revaluation or reprogramming of existing forms. AItlist turns that motivation into a field-aware diagnostic; it does not measure global novelty.
Public-reception resilience	AItlist-specific curatorial diagnostic; Eco as a boundary on interpretation	Proposal-level heuristic against a dossier being exhausted by a readily available generic framing. It is not measured audience reception, and Eco is used only to clarify that interpretive plurality can coexist with work-specific structure.
One dominant operation	No art-theoretical equivalence required	Explicit search representation and economy constraint used as a conservation target during development. It is not an ontological claim that a realized artwork has one objectively true operation.
Ethical/public-context diagnostics	Bishop; Kester; practical responsibility in commissioning and exhibition	Diagnostic-only reviewer outside the ten-dimensional score vector and winner ordering. Theory motivates keeping social, ethical, and artistic judgment distinguishable; AItlist does not reduce them to one common scalar.
Holistic external preference	Comparative LLM evaluation	Held-out outcome measure that deliberately does not reproduce the internal ten-dimensional scorecard. It is model preference over textual proposals, not artist/curator judgment or realized-artwork quality.

2.4 Why Proposal Improvement Is Non-Monotonic

AItlist is explicitly designed to improve proposals. The difficulty is not that artistic development lacks an evaluative objective altogether: in the experiments below, we can ask whether resulting proposals receive stronger independent holistic judgments. The difficulty is that open-ended artistic development offers no reliable local gradient toward that objective. A valid diagnosis does not uniquely determine a valid repair. A critique can correctly identify a weakness while the most direct correction removes or flattens another relation on which the proposal’s value depends.

This makes local and holistic improvement non-equivalent. A revision can become:

- more feasible while making the artistic operation less necessary;
- more immediately legible while closing a productive ambiguity;
- more interactive while weakening a precise spectator relation;
- more visually forceful while disconnecting the image from the material or spatial operation that gave it meaning.

We use *non-monotonic improvement* for this possibility: movement in an apparently desirable local direction does not guarantee movement toward the holistic objective. Review should therefore identify liabilities and candidate interventions without treating the obvious local fix as globally beneficial. In AItlist, review evidence is converted into a local search hypothesis rather than an automatic rewrite: challengers are generated from the unchanged incumbent, and the incumbent remains eligible to survive if no challenger earns replacement.

Two recurring cases make this trade-off concrete. First, clarity can conflict with productive ambiguity. Eco’s *Open Work* offers an influential account of artistic structures that permit multiple realizations or interpretations without collapsing into arbitrariness, while Gaver, Beaver, and Benford argue in design that ambiguity can function as a resource for interpretation rather than merely as missing information [38, 40]. The implication for AItlist is not that ambiguity is inherently valuable: incoherence and underspecification remain genuine weaknesses. The

narrower point is that a productive ambiguity can leave something unresolved while still organizing what a spectator can perceive, infer, or experience. An instruction such as “make this clearer” may remove accidental noise, but it may also close the unresolved relation that gives the proposal interpretive pressure.

Second, engagement can conflict with spectator specificity. Bishop and Ranciere give different reasons to reject a simple hierarchy in which converting spectators into visible actors automatically constitutes artistic progress [30, 29]. A proposal may indeed be weak because the spectator relation is generic or inert, but adding visitor input does not by itself repair that weakness. It can increase observable activity while replacing a precise relation of attention, distance, delay, access, or non-action with generic interaction. The relevant question is whether the intervention strengthens the proposal’s artistic operation and spectator relation, not whether the spectator is made to do more.

The experimental consequence is evaluated in RQ2 by comparing every adopted replacement with the exact parent state it displaced; Section 5.5 defines that estimand.

2.5 Maintaining a Sampled Possibility Set

The same uncertainty has a second consequence at the level of the search population. If a proposed repair is not guaranteed to improve its lineage, and different lineages can preserve different strengths, ideation should not collapse immediately to a single answer. Computational-creativity research describes creativity in terms of exploring and transforming spaces of possibilities rather than producing one isolated artifact [41, 42, 43]. Design-fixation research provides a complementary warning: early examples can constrain subsequent concepts even when alternatives are explicitly requested [44]. Large language models make the tension especially visible. Generating five fluent proposals does not guarantee five serious directions. A model can repeatedly vary the same metaphor, object type, interaction format, or rhetorical framing while changing vocabulary. For the intended use case, AArtist therefore keeps five proposal lineages alive and returns all five surviving dossiers to the artist. We call this finite population the sampled possibility set. The phrase is deliberately modest. Five proposals are not the possibility space of contemporary art, do not exhaust the generative model, and are not claimed to be the optimal search width. They are the alternatives the system has actually constructed and can therefore expose to a user and evaluate.

This population view makes distinctions observable that winner-only evaluation collapses. A system may:

- occasionally produce one outstanding proposal while leaving the rest of the set weak;
- improve several candidates simultaneously;
- narrow the set while increasing its measured quality;
- generate a strong option but rank it poorly;
- produce several incompatible but serious directions for later human selection.

The quality of the available set and the ability of the system to surface one member of that set are therefore different empirical objects.

Sampled-set quality is not diversity

Maintaining several candidates does not by itself imply a diverse candidate set. A pool can be broad but weak, while another can be narrow but strong; and diversity can refer to semantic difference, artistic operation, material strategy, visual form, spectator position, political stance, art-historical distance, or combinations of these. Parallel lineages and heterogeneous challenger roles are mechanisms intended to resist premature collapse, not empirical evidence that diversity actually increased.

AArtist therefore treats sampled-set quality and breadth as separate empirical objects. The primary evaluation asks whether the retained proposal set becomes stronger under holistic judgment. Latent-semantic breadth is measured separately and only post hoc. As the results later show, higher measured proposal quality need not coincide with greater semantic spread.

2.6 Scope of Computationalization and the Empirical Target

AArtist computationalizes proposal development at the level of textual commission hypotheses. Its internal ten-dimensional quality model is a bounded search surrogate used to structure

generation, critique, preservation, intervention, and ranking; it is not an account of artistic truth. The resulting dossiers are proxies for future installations, not fabrication specifications or evidence about realized works (see Section 10.2 for the text-to-work validity boundary).

The empirical outcome is deliberately separated from that surrogate. Primary evaluation uses a held-out GPT-5.6 Sol evaluator that compares complete anonymized proposals holistically rather than reproducing AItist’s ten-dimensional scorecard (see Section 5.3 for the judging protocol). The experiment therefore asks whether search guided by the internal surrogate transfers to a broader external model judgment rather than treating success under the surrogate as self-validation. That model preference is itself not a proxy for artist or curator judgment or realized-artwork quality (see Sections 10.1–10.2).

3 Related Work and Positioning

The term concept denotes different objects across the literatures adjacent to this work. In image generation, it can mean an object category or semantic attribute; in professional concept design, it often denotes a visual direction for a character, environment, or product. Here, concept development refers to the formation of the artistic proposition of a work: what the installation does, and how its material, spatial, spectator, historical, and institutional relations constitute that proposition. This distinction is important because recent AI systems already support sophisticated forms of ideation and creative generation, but usually operate on a different creative object.

We review three relevant lines of work. Contemporary-art practice shows generative systems moving upstream from artifact production toward artistic dialogue and the conception of possible works. Research on AI-assisted ideation establishes structured exploration and iterative development as computational problems in adjacent domains. Agentic systems further provide mature mechanisms for critique, branching, and selection. Together, these literatures motivate the task studied here while leaving the early development of a contemporary-artwork proposition largely unformalized as an evaluated computational problem.

3.1 AI in Contemporary-Art Practice

Contemporary-art practice has already begun to move generative AI upstream from the production of finished images into the formation of possible artworks. Agnieszka Kurant’s *Errorism* (2021) provides a particularly direct precedent. GPT-2 and GPT-3 were trained on descriptions of Kurant’s previous works and her essays and used to generate descriptions of conceptual works that she had never made but could potentially create; Kurant subsequently visualized this speculative material as holographic animations [3]. The significance of *Errorism* for the present work is not simply that an artist used a language model. The model operates at a moment when a future artwork is still being articulated: it extrapolates possible works from an existing artistic practice rather than merely rendering a specification supplied by the artist.

Other recent practices extend this relation between generative systems and artistic conception. In Alexander Reben’s *I’m Not A Painter* (2022), a custom neural network and GPT-3 generate descriptions of nonexistent artworks; Reben curates these outputs, passes selected descriptions through text-to-image systems such as DALL-E 2, and selects images for physical realization [45]. Jonas Lund describes the works in *In the Middle of Nowhere* (2023) as having been conceptualized and produced through extended conversations with ChatGPT [4]. Such projects make language models part of a process of proposing, negotiating, and selecting possible works, rather than using them solely as image-making tools.

By 2024, related practices were also appearing as persistent, artist-specific systems. Cai Guo-Qiang’s cAI is a custom multimodal model that learns from Cai’s artworks, archives, and interests and is positioned by his studio as a partner for dialogue and collaboration [5]. cAI participated in the conception and choreography of the Getty-commissioned *WE ARE* (2024) [6]. Anicka Yi’s *Emptiness* project similarly uses an algorithm trained on years of her studio’s output as a “digital twin” of her practice [46]. Holly Herndon and Mat Dryhurst’s *The Call* (2024–25) makes another aspect of this shift explicit by treating the construction of vocal datasets, governance frameworks, and choral AI models as part of artmaking itself [47].

These practices make the motivating use case for AItist substantially less speculative. Established artists are already allowing generative systems to propose possible works, participate in artistic direction, and develop persistent computational counterparts to their practices. At the same time, these are artist-specific artistic procedures. Their systems are embedded in a particular oeuvre, studio process, exhibition, or authored experiment; they are not intended to

accept an unfamiliar artistic problem and systematically develop several complete responses to it.

Recent empirical studies of artists working with generative AI reinforce this distinction. A preregistered experiment with professional visual artists shows that artistic expertise remains consequential when working with text-to-image AI [48], while qualitative work with contemporary artists characterizes human–AI co-creativity as a sustained, relational process shaped by artistic intuition, control, and adaptation to a system’s affordances [49]. Such findings motivate treating AIrtist as one bounded computational component of a larger artistic process: the artist supplies the problem and remains responsible for subsequent artistic development, while the present system investigates what computational support can contribute between those stages.

A small number of computational-art projects have also begun to place LLM agents inside contemporary-art discourse. Artism, for example, combines a simulated social network of virtual artist agents with a system that recombines fragments of contemporary-art discourse into synthetic movements, visualizations, and criticism. Its purpose is primarily the simulation and critique of patterned artistic production rather than development of an artist-authored project, and the reported evidence is a preliminary case study rather than a controlled test of artwork concept development [50].

3.2 AI-Assisted Ideation and Design-Space Exploration

Early-stage ideation is considerably more mature as a computational problem in adjacent creative and professional domains. AIdeation, for example, supports professional concept designers in film, television, and games through brainstorming, reference exploration, and recombination. Its development is grounded in a formative study with 12 professional designers and evaluated through a controlled study with 16 professionals and a subsequent deployment in four studios [7]. The work provides strong evidence that generative systems can usefully intervene before a visual design has stabilized rather than merely execute a completed specification.

CoExploreDS addresses a related property of ill-defined design problems: problem formulation and solution development can co-evolve. The system represents emerging problems and candidate solutions explicitly and uses multiple reasoning modes to propose further directions [51]. Its relevance to AIrtist lies in establishing structured exploration of an evolving creative state as a serious design problem in its own right.

Research ideation offers an autonomous analogue. ResearchAgent decomposes scientific ideation into literature-grounded problem identification, method and experiment generation, review, and iterative revision [8]. The AI co-scientist similarly maintains and develops multiple hypotheses through generation, reflection, ranking, evolution, and meta-review [9]. Si et al., comparing LLM-generated research ideas with ideas written and blindly reviewed by more than one hundred NLP researchers, further show that idea generation and idea evaluation need not improve together: their agent produces ideas judged more novel on average while exposing failures in self-evaluation and a lack of diversity [52].

These systems establish that AI-assisted ideation, structured exploration, multi-candidate development, and iterative criticism are not new in themselves. The distinction in AIrtist concerns the state being developed. Visual concept design ultimately produces production assets; product design converges toward functional solutions; research ideation produces hypotheses and experimental plans. In contemporary-art concept development, changing a material relation, spatial arrangement, spectator position, or institutional procedure can change the identity of the proposed work itself. This motivates the domain representation introduced in Section 2 rather than treating the task as generic brainstorming with an art-themed prompt.

A complementary body of work also cautions that generative assistance does not necessarily broaden exploration. Controlled studies report stronger fixation under exposure to generated visual examples [10], greater semantic similarity across LLM-assisted ideators [53], and improvements to individual outputs accompanied by reductions in collective diversity [11]. Recent work on multi-agent ideation likewise shows that denser interaction and larger groups can produce diminishing diversity returns and premature convergence rather than broader exploration [54]. These findings make the existence of multiple candidates a poor proxy for the quality or breadth of the resulting possibility field.

The Cultural Alien Sampler provides an art-generation-specific example of explicitly measuring such exploration. Its computational object, however, differs sharply from that studied here. CAS deliberately reduces its search space to combinations from a finite vocabulary of 3,500 visual concepts—examples include sequences such as *ukiyo-e + machinery + hunter*—which are subsequently converted into prompts and rendered by an image generator. Its relevance is

methodological rather than task-level: the paper shows that different generative policies exhibit measurably different repetition, exploration-radius, and saturation behavior over repeated generations [55].

This literature motivates two aspects of our evaluation. First, the retained candidate population should be treated as an object of analysis rather than assuming that producing several outputs constitutes meaningful exploration. Second, candidate quality and candidate breadth are distinct properties: a system can improve one without necessarily improving the other.

3.3 Agentic Revision and Selection

Iterative critique, role specialization, and selection are established components of current agentic systems. ResearchAgent uses collaborative reviewing agents before revising scientific ideas [8], while the AI co-scientist combines reflection and evolutionary development with tournament-based ranking among hypotheses [9]. More general inference-time methods such as Self-Refine and Tree of Thoughts replace a single completion with iterative feedback or exploration over multiple candidate reasoning states [56, 57].

Role-specialized refinement also appears in creative-generation systems. CREA, for example, coordinates Creative Director, Prompt Architect, Art Critic, and Refinement Strategist agents for creative image editing and generation. Its task remains explicitly image-level—the system transforms or generates objects such as couches, cars, and other visual categories—and its critic evaluates rendered images against generic creativity dimensions before further regeneration [58]. CREA therefore demonstrates the maturity of multi-agent creative refinement as a mechanism, rather than providing a precedent for the development of an artwork proposition.

Airtist builds on the general premise that additional inference can be allocated to critique and alternative generation, but changes the status of revision. In canonical iterative-refinement methods such as Self-Refine, feedback is used to produce a revised successor [56]; ResearchAgent likewise feeds reviewer feedback into iterative revision [8]. For the task studied here, identifying a weakness does not imply that the available repair is preferable to the existing proposal. Review is therefore first converted into an asymmetric local state: what gives the incumbent its strongest identity, what most needs intervention, which familiar operating logic should be left, and what forms of repair should not count as progress. Focused and Structural challengers are generated independently from the same parent, while the unchanged incumbent remains eligible.

The resulting empirical question differs from ordinary endpoint refinement: when the system actually elects to transform an artistic proposal, is the adopted intervention preferable to the exact state it was intended to improve? Exact-parent comparison makes that question directly observable.

Selection creates a second separable problem. Research-ideation results already suggest that generative ability can exceed reliable self-evaluation [52]. Airtist therefore does not infer ranking quality from proposal quality. Its terminal selectors act on exactly the same generated five-candidate set and are evaluated against an external utility unavailable during selection. This isolates failure to recognize a strong candidate from failure to generate one in the first place.

3.4 Positioning Airtist

The preceding literature leaves a specific task gap. Contemporary-art practice already demonstrates language models and other generative systems participating upstream of final realization, including the proposal of hypothetical artworks and ongoing artist-specific computational collaborators. Adjacent computational research already provides sophisticated methods for ideation, structured exploration, critique, revision, and ranking. Yet across the systems and documented artistic practices reviewed here, we did not identify an evaluated general-purpose system whose primary task is to take a previously unseen, under-specified artist-authored contemporary-art installation brief and develop several complete hypotheses for what the artwork itself could become.

To our knowledge, Airtist is the first evaluated computational system designed specifically for this task. Its outputs are not image concepts or thematic suggestions, but complete project hypotheses in which an artistic operation is articulated together with its material, spatial, spectator, field, and institutional realization.

Airtist further proposes a particular computational treatment of concept development. Complete project hypotheses persist as competing states; critique specifies both preservation and transformation constraints; proposed changes compete with the incumbent rather than auto-

matically replacing it; and evaluation distinguishes the quality of the resulting proposal set, the consequences of adopted interventions, and the effectiveness of terminal selection.

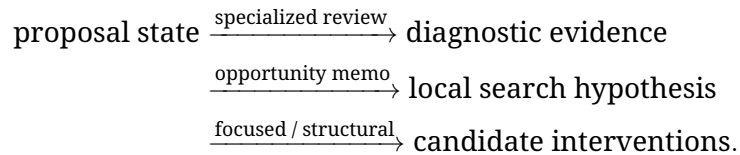
Contemporary-art installation ideation is therefore the substantive domain of the work, while also exposing a broader issue for open-ended creative agents. The next section gives the agentic computational formulation used to study this task.

4 AItist: An Agentic Architecture for Installation Concept Development

AItist takes a single artist-authored brief and returns five developed installation proposals in ranked order. In the evaluated configuration, the artist supplies no intermediate feedback: proposal construction, review, branching, competition, and ranking all occur internally after the initial brief is submitted. The five returned dossiers should therefore be understood as ranked starting points for subsequent artistic development, not as finished artworks.

AItist represents concept development through persistent proposal lineages rather than as a chain of increasingly polished rewrites. Computationally, those lineages induce a local-search process over complete proposal states: each maintains an incumbent proposal, constructs diagnostic evidence about it, converts that evidence into a constrained local search hypothesis, generates two differently instructed challengers from the unchanged parent, and allows the incumbent to survive if neither challenger earns replacement. A terminal ranker operates only after the five lineages have been developed.

The architecture therefore separates three kinds of state that generic “critique and rewrite” systems often collapse:



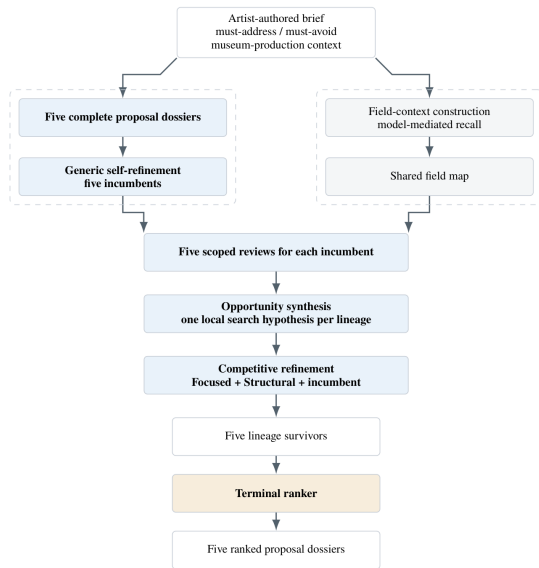
A review is not itself a revision instruction. An opportunity memo is not a revised proposal. And a generated challenger does not automatically replace its parent.

Figure 1 summarizes the end-to-end architecture. Proposal construction and field-context construction begin as separate streams. They first meet at specialized review and opportunity synthesis. Competitive refinement then proceeds independently across five lineages before a separate terminal ranking stage orders the five surviving dossiers. Table 4 summarizes the interfaces. “Output” denotes the principal object passed forward rather than every piece of recorded provenance or diagnostic metadata.

Table 4: Stage interfaces in AItist.

Stage	Principal input	Principal output
Initial proposal construction	brief, explicit constraints, museum/production context, internal proposal-quality surrogate	five complete installation dossiers
Generic self-refinement	one initial dossier, brief and constraints, shared quality guidance	five developed incumbents
Field-context construction	brief and explicit constraints	shared field map
Specialized review	incumbent dossier, brief, field map, reviewer-specific scope	five scoped diagnostic packets
Opportunity synthesis	immutable incumbent, review packets, field evidence	compact local search hypothesis
Challenger generation	immutable incumbent, brief, opportunity memo	focused reconstruction + structural escape
Challenger eligibility	each new challenger, brief and hard constraints	gate eligibility decision
Lane comparison	incumbent + eligible challenger(s)	one surviving dossier per lineage
Terminal ranking	five survivors, brief and constraints	total ordering of all five dossiers

(a) End-to-end system



(b) One lineage expanded

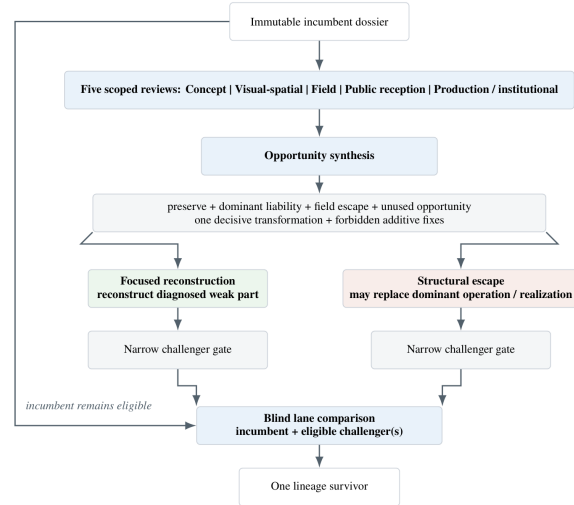


Figure 1: AArtist architecture. (a) End-to-end flow from one artist-authored brief to five ranked proposal dossiers. Proposal-state construction and field-context construction are initially separate; the shared field map enters specialized review and opportunity synthesis. (b) One lineage expanded. Reviews produce diagnostic evidence; opportunity synthesis compresses it into one local search hypothesis; Focused and Structural challengers are independently generated from the unchanged parent. Only challengers are gated. The incumbent bypasses the gate, remains eligible, and competes with all eligible challengers in the blind lane comparison. The terminal ranker is a separate stage over the five lineage survivors.

4.1 Input and Persistent Proposal State

4.1.1 Artist brief and commissioning context

The input to AArtist is a partially specified artistic problem rather than a request for a finished artwork. The intended user is an artist who has a rough direction, knows some conditions that matter, and may know several solutions that should explicitly be avoided, but does not yet know what form the installation should take. AArtist represents this input through three elements:

- a free-text project brief;
- explicit must-address conditions;
- explicit must-avoid conditions.

The free-text component states the artistic or situational problem. Must-address conditions identify pressures that should remain active rather than being replaced by an easier neighboring theme. Must-avoid conditions remove unwanted solution families without prescribing what should replace them. A prohibition on cameras, for example, removes one realization mechanism; it does not specify what a non-camera installation should be.

Unless the brief specifies another context, AArtist assumes a temporary or medium-duration contemporary-art museum commission. The proposal may occupy a gallery, foyer, threshold, circulation area, stairwell, entrance zone, or service-adjacent museum space, but it is assumed to remain within a museum-controlled footprint. An unspecified brief is not silently reinterpreted as a city-street, municipal-infrastructure, or other civic project outside ordinary museum authority. The default production context assumes an approximate USD 150,000 project envelope covering project-specific fabrication, materials and equipment, specialist engineering and temporary technical labor, installation and deinstallation, safety measures, maintenance requirements, project-specific staffing, and contingency. As Section 2 emphasized, this is a reasoning context for a textual commission hypothesis, not evidence that engineering or fabrication has been validated.

Together these elements define the search problem: what artistic pressure must remain active, which solutions are inadmissible, and within what institutional world must a proposal function? They do not prescribe a medium, object type, interaction model, or artistic style.

Table 5: Exact fields of the installation dossier.

Dossier field	What it represents
title	A working identifier for the proposal. It may contribute to framing but is not itself treated as an artistic operation or quality dimension.
artistic_thesis	What the work proposes, tests, places under pressure, or asks us to encounter. In Section 2 terminology, the dossier’s concise articulation of artistic stakes.
artistic_operation	The dominant action or dependency through which the installation enacts the thesis rather than merely stating it.
concept_description	An integrative description connecting thesis, operation, visual/spatial/material realization, spectator relation, and institutional situation.
visual_logic	What is concretely visible or perceptible: dominant image or material presence, scale, silhouette, surface, light, color, texture, and viewing conditions.
spatial_logic	How route, position, access, distance, sequence, threshold, orientation, concealment, visibility, waiting, and bodily movement organize the installation.
material_logic	What materials, media, processes, absences, or realization rules physically, temporally, historically, socially, or institutionally do; which properties are constitutive and which may vary.
spectator_encounter	The spectator’s position, attention, movement, action or non-action, access, duration, visibility, knowledge, and consequence within the installation.
difference_from_field	The proposal’s own hypothesis about how it differs from nearby artistic formats or precedents; not independent evidence of art-historical novelty.
risks_and_failure_modes	The proposal’s own account of how the work could become generic, decorative, gimmicky, incoherent, didactic, technically fragile, or otherwise lose artistic pressure.
institutional_feasibility	Site, staffing, maintenance, accessibility, safety, permissions, liability, equipment, construction, and production dependencies required for the commission hypothesis to operate.

4.1.2 The installation dossier as persistent authored state

The object manipulated by AArtist is a structured installation dossier. It is the persistent authored state of a proposal lineage. Reviews, field evidence, and opportunity memos are attached to or derived from that state; they do not silently rewrite it. Challengers are generated from an immutable copy of the incumbent, and any selected challenger receives a traceable parent relation. This separation is what later allows the experiment to compare a selected intervention against the exact textual parent it displaced.

The implementation uses the following authored fields. To avoid ambiguity, we use the exact field names consistently throughout the paper.

Several distinctions in this representation matter downstream. `difference_from_field` is candidate-authored positioning. It should not be confused with the independently constructed field context described in Section 4.2. `risks_and_failure_modes` is likewise the proposal’s own anticipation of failure. It should not be confused with the later independent reviewer failures. And `institutional_feasibility` represents the dossier’s production hypothesis. It is later pressure-tested by a separate Production/Institutional reviewer. Section 2 defines the conceptual role of artistic operation and the relationship among stakes, operation, and realization. The dossier makes those distinctions explicit enough for later agents to diagnose them separately. A compelling thesis can coexist with arbitrary spatial logic; a visually specific proposal can use interchangeable material; a materially interesting object can establish no precise spectator relation. Keeping these fields separate allows the system to ask what should remain invariant and what can be reconstructed.

4.2 Initializing Search: Five Proposal Lineages and a Shared Field Context

AArtist prepares two inputs for specialized development: a five-proposal candidate population and a shared hypothesis about the artistic field surrounding the brief. Both derive from the brief, but the streams are initially separate. Field context does not determine which five initial proposals are generated; it enters their development later, through review and opportunity synthesis.

4.2.1 Five proposal lineages

The first stream produces five complete installation dossiers in one batched generation. Each should constitute a distinct proposal rather than a continuation or paraphrase of another. Each dossier then undergoes one generic self-refinement pass. The pass may revise any authored dossier field while preserving the underlying proposal and strengthening brief fidelity, operation-realization coherence, spatial and material specificity, spectator encounter, and plausible museum realization. Initial proposal construction and this generic self-refinement already operate under installation-specific quality guidance. They do not yet receive the synthesized field map or specialized review packets introduced below.

All five self-refined proposals are retained. They become five incumbents, one per persistent lineage. AArtist therefore does not select a single concept and recursively improve it. It begins

specialized search with five complete hypotheses whose histories remain independent.

4.2.2 Field-context construction

In parallel, AItist constructs contextual evidence intended to help later stages recognize familiar artistic attractors and possible routes away from them. This stage is model-mediated recall, not provenance-bearing retrieval. It does not search a verified external bibliography. Recalled artworks, artists, critical writing, and neighboring cultural or material references serve as associative context for search and must not be interpreted as factual evidence that a proposal is globally novel in contemporary art.

The recall process considers four complementary kinds of material:

1. concrete artworks;
2. artists and collectives;
3. art writing, including criticism, catalogues, interviews, and exhibition discourse;
4. relevant material outside art, including cultural, historical, spatial, legal, technical, or social contexts.

Relevance is deliberately broader than topical similarity. A reference may matter because of material behavior, image logic, object presence, atmosphere, spectator position, temporal structure, institutional relation, bodily scale, sound, stillness, fiction, or another transferable pressure.

The field synthesizer converts this material into five explicit output categories:

- `nearby_lines` – nearby artistic approaches or operating logics;
- `overused_formats` – familiar uses that may become generic if repeated mechanically;
- `productive_differences` – dimensions along which the proposal could depart meaningfully;
- `taboo_zones` – locally dangerous attractors or especially obvious forms of repetition;
- `field_insights` – more specific observations grounded in the recalled material.

These are working search hypotheses. `overused_formats` does not prohibit a medium, `taboo_zones` does not create new must-avoid constraints, and `field_insights` does not certify art-historical facts. The same synthesized field context is shared across all five lineages.

4.3 Specialized Multi-Agent Review: Diagnosis as Evidence

An installation dossier can fail in several independent ways. A compelling proposition may remain spatially arbitrary. A precise spatial situation may rely on an interchangeable material. An otherwise coherent work may reproduce a familiar artistic format. Its visible public signal may collapse into a gadget or awareness display. Or its operation may depend on hidden staffing, inaccessible circulation, or an unsafe physical mechanism. AItist therefore reviews each incumbent through five deliberately scoped diagnostic agents. The split is not intended to simulate a curatorial committee voting on artistic value. It decomposes AItist’s ten-dimensional internal quality surrogate from Section 2 into non-overlapping diagnostic ownership.

Table 6: Review agents and ownership of the internal surrogate.

Reviewer	Owned internal dimensions	Primary diagnostic question
Concept	raw-prompt fidelity; core-tension enactment; artistic-operation necessity; material necessity	Does the proposal enact the brief through one necessary dominant operation, or could its form be replaced without substantial artistic loss?
Visual-spatial	spatial encounter; visual encounter; spectator specificity	What does a spectator concretely see, occupy, cross, wait for, avoid, or experience, and how do position and duration change that encounter?
Field	field distinctiveness	Under the model’s recalled field context, does the proposal transform a nearby artistic logic or merely reproduce an available format?
Public reception	public-reception resilience	What plausible public reading could flatten the proposal into a gimmick, spectacle, awareness device, wellness environment, decorative object, or generic engagement format?
Production / institutional	production/institutional viability	Can the stated artistic relation plausibly operate with respect to fabrication, safety, accessibility, maintenance, staffing, permissions, liability, and the shared production envelope?

4.3.1 Review packet interface

Each reviewer receives:

- the incumbent dossier;
- the original brief and explicit constraints;
- relevant shared field evidence;
- reviewer-specific scope instructions.

It returns diagnostic evidence rather than a rewritten proposal.

A review packet contains:

- strengths;
- failures;
- revision_demands;
- diagnostic severity;
- diagnostic repairability;
- diagnostic confidence;
- scope_handoff_notes when a concern primarily belongs to another reviewer.

The Production/Institutional reviewer additionally returns a structured production-envelope assessment when enough information is present.

The numerical diagnostics are not votes. They are not averaged across reviewers to decide whether a proposal survives. In the frozen review contract, failures and revision demands are explicitly the primary review output; severity, repairability, and confidence summarize the diagnosis within that reviewer's restricted scope. An empty failure list is valid. The stage can therefore be written as:

$$D_i \xrightarrow{\text{five scoped reviewers}} R_i = \{r_i^{\text{concept}}, r_i^{\text{visual}}, r_i^{\text{field}}, r_i^{\text{reception}}, r_i^{\text{production}}\}.$$

The result is evidence about a proposal, not five competing rewrites.

Running example: Delayed Projection (Brief 035) The brief asks for a work about public shame while excluding cameras, mirrors, humiliation performance, and moralizing text. The incumbent Delayed Projection nevertheless depends on live capture: infrared cameras record visitors moving through the gallery, a media server holds the footage for ten minutes, and processed silhouettes are then projected onto a semi-transparent scrim across the room. A new visitor therefore encounters traces of people who occupied the same space earlier but are no longer physically present.

The reviewers separate this valuable temporal relation from the mechanism that produces it. The Concept reviewer identifies the delayed, absent audience as a genuine strength but flags the cameras as a direct must-avoid violation. The Visual-Spatial reviewer treats the delayed silhouettes and scrim as a strong perceptual-spatial situation and records no in-scope failure. The Field reviewer likewise regards temporal displacement as meaningful while handing the camera prohibition back to the appropriate scope. The Public-Reception reviewer warns that the projection could collapse into a technological gimmick or photographic backdrop. The Production / Institutional reviewer identifies privacy and consent liabilities and proposes administrative mitigations such as signage, opt-out procedures, and a data-retention policy. Those additions would leave the prohibited recording mechanism intact. This disagreement is exactly why review is separated from revision: opportunity synthesis can ask whether the experience of an absent public can survive while live capture is removed, rather than concatenating every local repair into a longer dossier.

4.4 Opportunity Synthesis: From Diagnosis to a Local Search Hypothesis

The opportunity-synthesis agent receives the immutable incumbent dossier, its five review packets, and the shared field context. It does not rank candidates and does not write a revised proposal. Its function is to transform heterogeneous diagnosis into one tractable search hypothesis. We use *opportunity state* for this conceptual local-search state and *opportunity memo* for the concrete recorded schema through which that state is passed to downstream challenger generation. This stage is deliberately lossy. Specialized reviews may expose several valid weaknesses, possible repairs, production cautions, and field similarities. Attempting to satisfy all of them simultaneously would encourage additive revision. Opportunity synthesis instead decides which problem should organize the next search move and what must survive while that problem is addressed.

The actual memo schema contains six substantive fields plus evidence references.

Table 7: Opportunity memo schema. Formal field names are paired with plain-language roles.

Exact field	Plain-language role
incumbent_strength_to_preserve	Preserve: the strongest property that should not be accidentally destroyed by the next intervention.
dominant_artistic_liability	Liability: the weakness that should organize the local search step.
field_similarity_to_escape	Field escape: a nearby operating format or artistic attractor that the next move should avoid reproducing mechanically.
unused_material_or_spatial_opportunity	Opportunity: an underused material or spatial degree of freedom available to the proposal.
one_decisive_transformation	Search anchor: one concrete transformation worth testing rather than a list of independent repairs.
forbidden_additive_fixes	Do not substitute: additions that should not be accepted as a substitute for addressing the central liability.
evidence_ids	Traceability back to the diagnostic evidence used in synthesis.

The phrase one decisive transformation can sound as if synthesis has already decided how the proposal must be rewritten. That is not its computational role. The memo proposes a shared intervention hypothesis that both challenger agents will test under different search permissions. It is not an accepted revision. The stage therefore answers: What should the next candidate try to change, and what should it protect while doing so? It does not answer: Which new proposal should replace the incumbent?

4.4.1 Brief 035 opportunity state

The actual memo for Delayed Projection is especially informative:

Table 8: Opportunity state for the Brief 035 running example.

Memo field	Recorded state
incumbent_strength_to_preserve	temporal delay that produces a ghostly, absent audience and makes withholding of presence felt
dominant_artistic_liability	live infrared cameras violate the surveillance prohibition and create privacy liability
field_similarity_to_escape	familiar recorded-surveillance installation logic
unused_material_or_spatial_opportunity	a floor fissure or material rupture that can make exclusion bodily without recording visitors
one_decisive_transformation	replace live capture with a pre-programmed generative silhouette loop while preserving temporal delay
forbidden_additive_fixes	mirrors, live camera feeds, moralizing explanation, literal waiting line

The memo resolves the reviewer conflict by preserving temporal delay while treating recording itself—not its administration—as the search problem. It records a search hypothesis rather than a validated replacement: preserve delayed absent presence, remove live visitor capture, and let challenger generation test a non-recording realization. This illustrates why review packets are evidence rather than revision instructions.

4.5 Competitive Refinement

For each lineage i , Airtist generates two challengers independently from the same immutable incumbent x_i and the same opportunity memo m_i :

$$L_i = \{x_i, c_i^{\text{focus}}, c_i^{\text{escape}}\}.$$

4.5.1 Focused reconstruction versus structural escape

The distinction between the two challenger roles is procedural rather than categorical. The frozen prompts define them as follows. A focused reconstruction:

- preserves the artistic problem and the memo-protected strongest element;
- is organized around the diagnosed weak component;
- may deeply reconstruct the weak artistic operation, material logic, spatial logic, visual logic, or spectator relation;
- attempts to remove the dominant liability rather than compensate for it through explanatory or administrative additions.

A structural escape:

- preserves the brief, artistic problem, hard constraints, and memo-protected strength;
- is explicitly permitted to replace the dominant artistic operation or realization when remaining close to the incumbent would preserve the diagnosed field similarity;
- must be materially different from the incumbent.

This means that a focused reconstruction is not forbidden from changing artistic operation. The distinction is instead one of search intent and permission: Search permission, not edit size. Focused: reconstruct the diagnosed weak part while remaining organized around the incumbent’s protected identity. Structural: if local reconstruction would preserve the same problematic artistic logic, permission extends to replacing how the proposal works as a whole. The difference resembles refactoring versus substitution, but even that analogy should not be taken too literally. The outputs are generated proposals, not deterministic transformations with formally guaranteed edit radii.

4.5.2 Same parent, same memo: what “reconstruct” and “replace” look like

Brief 044 provides a useful controlled example because both challenger agents receive the same parent and same opportunity memo. Weighted Light Path uses twelve visitor-activated pressure sensors aligned with twelve wall elements. Activating a sensor starts a fixed delay—different across positions—after which the corresponding LED ribbon becomes visible: visitor action → fixed delay → visible response. The opportunity memo says to preserve that fixed hierarchy of waiting while escaping the generic light-up response. Exact sensor placement is a production choice rather than part of the artistic distinction being tested.

The two challenger modes respond differently. Focused reconstruction – Delayed Glow Wall. The focused branch keeps the same operative grammar: pressure → fixed delay → light becomes visible. Instead of an LED ribbon switching on, a brief UV pulse excites a phosphorescent strip, which then glows for several seconds. The unequal delay still culminates in light, but the visible event and material process are rebuilt.

Structural escape – Delayed Reveal Wall. The structural branch preserves the same trigger and delay hierarchy but abandons light as the dominant response: pressure → fixed delay → mechanical reveal. After the programmed latency, a low-voltage gear motor releases a small recessed panel and exposes a colored surface.

Table 9: Same parent and memo, different transformation permissions in Brief 044.

	Focused	Structural
Trigger	pressure	pressure
Core temporal problem	unequal fixed delay preserved	unequal fixed delay preserved
Dominant event	reconstructed light event: phosphorescent afterglow	replaced event: mechanical reveal
Search move	rebuild weak realization from inside	leave the lighting realization altogether

The example is about search permission rather than demonstrated artistic superiority. Focused reconstruction rebuilds the weak realization from inside the existing light-response grammar; Structural escape is allowed to leave that grammar altogether. Neither branch depends on an unusual physical effect, and the example does not claim that the mechanical reveal is artistically better than the phosphorescent one.

4.5.3 The distinction is a permission, not a guarantee

The roles are not guaranteed to produce different semantic distances. In Brief 030, for example, both agents independently move away from the clock-mediated parent toward almost the same physical solution family: reclaimed bricks of unequal mass pull different points of one shared veil through cables and pulleys. When the brief and opportunity memo strongly constrain a plausible repair, different search permissions can therefore converge on the same underlying causal move. Structural escape is permission to replace more, not a requirement to produce maximal novelty.

4.5.4 Challenger eligibility

Generating a challenger does not make it a legitimate competitor. Each new challenger first passes a narrow eligibility gate. The incumbent is not re-gated and remains available throughout the lane. Deterministic checks remove exact duplication of the parent or sibling challenger. The model-based gate is reserved for narrow invalidity conditions such as:

- malformed output;
- negligible material difference from the incumbent;
- explicit must-address or must-avoid failure;
- loss of brief fidelity;
- another concrete hard-constraint violation;
- severe safety failure.

Ordinary artistic weakness, unfamiliarity, production complexity, cost within the broader development envelope, or incomplete readiness are not by themselves hard-gate conditions. This distinction connects directly to Section 2: the gate defines eligibility; it is not another component of the ten-dimensional internal quality surrogate. A proposal may be eligible and still artistically weak. Such a candidate should lose through comparison rather than disappear through a disguised aesthetic veto.

4.5.5 Blind lane competition

The incumbent and all gate-eligible challengers then enter lane competition. Candidate origin is hidden from the internal selector. The comparison sees authored dossier content, the brief and explicit constraints, and AIrtist’s ten-dimensional internal surrogate of proposal quality. It does not receive reviewer packets, opportunity memo, candidate role, lineage history, or prior selection outcome. This prevents the lane selector from rewarding a challenger merely because earlier agents invested effort in producing it.

The update has three cases.

1. No challenger passes the gate. The incumbent survives automatically.
2. Exactly one challenger passes. Incumbent and challenger receive independent pointwise assessment under the ten-dimensional internal surrogate. The higher-scoring proposal survives, with deterministic resolution of an exact tie.
3. Both challengers pass. All three candidates receive pointwise assessment and are additionally ranked under three cyclic presentation orders so that each candidate occupies each list position once. These rankings are aggregated through Borda evidence and first-place counts, followed by pointwise score and a deterministic final tie-break.

No incumbent bonus is used. Thus preservation is possible without being privileged. One proposal survives each lineage. Repeating the procedure independently across five incumbents produces the five-proposal set passed to terminal ranking. The distinction between this internal decision and the paper’s external evaluation is important. The lane selector acts under AIrtist’s internal ten-dimensional surrogate. The later GPT-5.6 Sol outcome used in the experiment is the held-out holistic comparison introduced in Section 2.6 and is unavailable to the search system.

4.6 Terminal Ranking

Competitive refinement answers: Which state should survive within each proposal lineage? It does not answer: In what order should the artist inspect the five surviving proposals? Airtist therefore uses a separate terminal ranker.

This distinction matters because the final interface exposes all five dossiers but does not expose them with equal salience. Rank 1 acts as the system’s strongest recommendation and can frame the artist’s first impression of the candidate set. The terminal ranker receives the five survivor dossiers, the original brief, and explicit constraints. It is blind to search history. It does not receive reviewer packets, field context, opportunity memos, incumbent/challenger roles, lane-selection decisions, or earlier internal scores.

4.6.1 Pointwise screening

All five proposals are first assessed independently under the same ten-dimensional internal surrogate used for proposal-quality comparison elsewhere in Airtist. The assessment yields ten quality scores, a decisive strength, a decisive liability, confidence, and readiness risks.

This stage also supports a narrow terminal hard veto, distinct from ordinary low quality. The veto is restricted to conditions such as severe explicit-constraint violation, physically impossible central mechanism, unsafe failure required for the work to function, or a safety defect that cannot be removed without replacing the central operation. Ordinary production difficulty, maintenance burden, staffing, familiarity, or a safely remediable accessibility problem do not automatically trigger a veto. A vetoed proposal is not deleted from the five-proposal output. It remains in the final ordering below admissible candidates and cannot be promoted into the finalist tournament. Terminal eligibility therefore affects rankability at the top of the list, not candidate-set size.

4.6.2 Top-of-pool tournament

Up to the three highest-scoring non-veto proposals become finalists. If fewer than three admissible candidates exist, the tournament shrinks rather than promoting a vetoed candidate simply to fill a fixed shortlist. Every finalist pair is judged twice with exact presentation reversal. For finalists a and b , one comparison presents a as Candidate A and b as Candidate B; the second presents the same dossiers in the opposite order. The pairwise comparator receives the brief, explicit constraints, and the two authored dossiers. It does not receive the candidates’ pointwise scores or search history. The two orientations are mapped back to candidate identity before aggregation.

Across finalist pairs:

1. logical wins and losses determine Copeland score;
2. accumulated orientation-level pairwise points provide the next ordering signal;
3. pointwise quality score breaks remaining ties;
4. a deterministic content-based rule resolves an exact residual tie.

Candidates outside the finalist set retain their pointwise order below the tournament finalists; vetoed proposals remain below admissible proposals. The final product is therefore a total ordering of all five surviving dossiers, not a shortlist and not a single surviving concept.

4.6.3 Why use a tournament?

The motivation is not that pairwise comparison is assumed to be intrinsically superior. Pointwise screening asks a broad question independently of the other candidates: How strong is this proposal under the internal surrogate? The tournament concentrates additional inference on a narrower question: When the strongest-looking candidates are compared directly, does their relative order change? This architecture was designed to concentrate expensive comparative judgment near the top of the fixed pool.

Whether that additional comparison actually improves the external quality of the first-ranked proposal is not assumed by the architecture. RQ3 evaluates exactly that question on identical generated candidate sets. The 50-brief experiment therefore treats the tournament as a tested ranking policy, not as an architectural component whose superiority is taken for granted.

4.6.4 System output

The final output presented to the artist is a ranked set of five complete proposal dossiers. The ranking is best interpreted as an allocation of attention among starting points, not as a claim that the first dossier constitutes the artwork or that lower-ranked proposals have no value. The architecture consequently preserves two objects at the end of search:

1. a fixed, traceable population of five developed installation hypotheses;
2. a separate internal policy for ordering that population.

This separation lets the evaluation distinguish the developed proposal population from the policy that orders it, without treating the top-ranked dossier as the only system output.

5 Experimental Setup

The experiment separates three empirical objects: RQ1 evaluates the retained five-proposal set, RQ2 evaluates adopted interventions against their exact parents, and RQ3 evaluates terminal ranking while holding all proposal texts fixed. A separate selected-output campaign provides a secondary end-to-end check.

Table 10: Evaluation decomposition. The rows isolate candidate-set quality, intervention quality, and ranking quality rather than treating the final top-ranked proposal as a single undifferentiated outcome.

Analysis	Empirical object	Held fixed	What changes	Main outcome
RQ1: sampled-set quality	Five retained proposals	Brief, generation model, external judging protocol	Search condition	Complete-pool preference / centered effect
RQ2: competitive intervention	Individual proposal lineage	Exact incumbent parent	Whether and how the lineage is transformed	Selected replacement vs. exact parent
RQ3: terminal selection	The same five A5 proposals	Every proposal text	Terminal ranking policy only	External Utility@1
Secondary end-to-end check	One surfaced proposal	Brief and holistic judging protocol	System whose selected output is shown	Holistic pairwise preference

5.1 Benchmark

Held-out commission briefs The benchmark contains 75 commission-style installation briefs. Briefs 001–025 were available during system development and are excluded from every reported evaluation. The empirical study uses the remaining 50 held-out briefs, IDs 026–075. The released benchmark is the frozen `benchmark_75_v1` dataset; the reported held-out slice contains ten briefs in each of five coverage categories.

Each brief represents a partially specified artistic problem rather than a request to reproduce an existing artwork or optimize toward a reference answer. As described in Section 4, it contains three kinds of information: a free-text project premise, a must-address field that identifies pressures the proposal should preserve, and a must-avoid field that removes unwanted solution families. Unless the brief itself specifies another setting, proposals inherit the common contemporary-art commission and production context defined in Section 4.

The five coverage categories are sampling devices rather than a taxonomy of contemporary-art practice, and the five-proposal population is an experimental design choice that also permits exhaustive cross-pool evaluation. See Section 2.5 for the sampled-set interpretation and Section 10.2 for benchmark and ecological-validity boundaries.

5.2 Experimental Conditions

All core generation and internal decision conditions use `openai/gpt-oss-120b` through OpenRouter, routed exclusively to Groq with provider fallback disabled. The frozen run uses temperature 0.8 across model-mediated stages. An experiment-level random seed of 42 controls deterministic ordering and related reproducibility mechanisms where the implementation uses a seed; it should not be interpreted as making stochastic model sampling deterministic. Candidate generation and all internal system decisions are frozen before external judging begins. The conditions are not four unrelated generations. They form a deliberately nested control structure.

Table 11: Held-out benchmark coverage. Categories are sampling devices, not proposed artistic genres. Examples reproduce the frozen benchmark semantics in shortened form.

Coverage category	What it stresses	Representative held-out example
Thematic / conceptual	Translating an abstract or social pressure into an operative installation rather than an illustration	Brief 026: public apologies continuously revised before they can be heard; revision must become a shared material burden rather than a written statement.
Site-specific	Making an existing spatial condition constitutive while preserving its practical use	Brief 037: unequal public ascent through an underused municipal stairwell, using handrails, landings, and daily wear while preserving every safety requirement.
Material-constrained	Producing an operation from a restricted material vocabulary	Brief 038: disputed land boundaries using only chalk, reclaimed brick, and rainwater-safe sealant; the work must remain readable after partial weathering.
Institutional / production-constrained	Working within access, staffing, maintenance, or public-use conditions	Brief 044: unequal waiting in a museum foyer during school visits; the work must tolerate touching and require no attendant beyond normal front-of-house staff.
Anti-cliche / difficult	Blocking highly available first-association solutions	Brief 035: public shame while excluding mirrors, cameras, humiliation performance, and moralizing text, with a precise passerby relation to what is withheld.

Raw five-proposal pool (B5-raw) B5-raw is the five-dossier population produced by the initial batched generation, before any self-refinement or specialized search.

Generic self-refinement (B5-self) Each of the five raw dossiers then undergoes the same single generic self-refinement pass used to initialize AItist. The resulting five proposals form B5-self. This condition controls for an important simple alternative: perhaps most of the benefit attributed to agentic search can be obtained by asking the same model to inspect and improve each initial proposal once. B5-self is also more than a baseline: these five exact proposal texts become the persistent incumbents from which both AItist and the matched iterative control begin. Consequently, the later A5–M5 contrast is not confounded by one system receiving a luckier initial sample. Both start from the same five self-refined parent dossiers with the same lineage identities.

Matched generic iteration (M5) M5 is the central mechanism control. Starting independently from each of the same five B5-self parents, it maintains five proposal lineages, generates two challengers per lineage, leaves the incumbent eligible, applies the same narrow challenger gate, and uses the same blind lane-comparison procedure as AItist. What it removes is the domain-structured organization of local search. M5 does not receive AItist’s specialized multi-role review, field-conditioned opportunity state, explicit preserve/liability/escape representation, or differentiated focused-reconstruction and structural-escape roles. Its two challengers instead perform generic proposal development. M5 therefore tests a stronger explanation than “AItist makes more calls.” Both systems perform multi-candidate local iteration from identical parents and select among old and new states. A residual A5–M5 difference is consistent with a contribution from the package of domain-structured search mechanisms rather than from additional sampling alone. It is important, however, not to overstate this control. M5 is substantially matched, not compute-identical. Exact prompt lengths, contextual inputs, call counts, and token expenditure are not equal. The A5–M5 contrast therefore does not estimate improvement per token or per model call, and it cannot identify which reviewer, opportunity-state field, preservation constraint, or challenger role is individually necessary or sufficient.

AItist retained set (A5) AItist begins from the same B5-self parents and applies the architecture described in Section 4: field-context construction, five specialized reviews per lineage, opportunity synthesis, independent focused and structural challengers, challenger gating, and competitive lane selection. One proposal state survives each lineage, producing a five-proposal set denoted A5.

Table 12: Nested control structure for the candidate-set experiment. A check mark means that the condition includes the corresponding property.

Experimental property	B5-raw	B5-self	M5	A5
Five complete initial dossiers	✓	✓	✓	✓
One generic self-refinement pass per dossier	–	✓	✓	✓
Same five B5-self parent texts	–	–	✓	✓
Five persistent lineages	–	–	✓	✓
Two challengers per lineage	–	–	✓	✓
Incumbent remains eligible	–	–	✓	✓
Same challenger gate	–	–	✓	✓
Same blind lane-comparison procedure	–	–	✓	✓
Specialized multi-role review	–	–	–	✓
Field-conditioned opportunity synthesis	–	–	–	✓
Explicit preserve / liability / escape state	–	–	–	✓
Focused and structural challenger roles	–	–	–	✓

5.3 Blind External Evaluation

Evaluation instrument The primary external evaluator is GPT-5.6 Sol, used in fresh ChatGPT Temporary Chats. It is separate from the gpt-oss-120b model used for generation and internal system decisions. The evaluator is instructed not to browse the web, use prior project context, or infer which system produced either candidate. For each comparison, the evaluator receives the original brief, its explicit must-address and must-avoid conditions, the common museum-commission context, and two complete anonymized proposal dossiers labelled Candidate A and Candidate B. The commission context assumes a temporary or medium-duration contemporary-art museum commission with an approximate production envelope of USD 150,000. The envelope covers project-specific fabrication, materials, equipment, engineering, temporary technical labor, handling, installation/deinstallation, permits, safety measures, consumables, maintenance materials, project-specific labor, and reasonable contingency; it excludes artist fee, international transportation, taxes, and ordinary permanent museum guards.

The external judging protocol is deliberately holistic. It asks which candidate is, overall, the more compelling, distinctive, coherent, visually and spatially convincing, responsive to the brief, and plausibly realizable contemporary museum installation. A tie is explicitly allowed. The evaluator is instructed not to reward length, fluency, polish, explicit theory, or self-criticism by themselves, and not to mechanically score dimensions. This distinction matters because the internal system and terminal rankers use the ten-dimensional bounded working model introduced in Section 2, whereas the external Sol protocol is not a copy of that scoring sheet. The two instruments overlap in the artistic concerns they attend to, but external pairwise utility is derived from a separate holistic judgment task rather than from AItist’s own internal component scores. The external evaluator does not receive system identity, source condition, proposal lineage, incumbent or challenger status, internal scores, internal rank, reviewer output, opportunity state, previous judging results, or the result of the opposite presentation orientation. External candidate utilities computed from these judgments are never returned to the generator, lane selector, or terminal rankers.

Complete cross-pool matrices For every brief, each of the five A5 candidates is compared with every candidate in each control pool. Thus A5 versus one control contributes $5 \times 5 = 25$ logical candidate comparisons per brief. Across 50 briefs and three control pools, $50 \times 3 \times 25 = 3,750$ logical comparisons are evaluated. This exhaustive design is central to RQ1. Evaluating only the top output of each system would entangle candidate generation with internal ranking and would make a strong system appear weak if its best candidate happened to be ranked poorly. The complete matrix instead makes the measured quality of every candidate observable independently of the system’s own selector.

Exact A/B reversal Every logical comparison is presented twice with exact candidate-order reversal. If one presentation shows proposal x as Candidate A and y as Candidate B, the paired presentation shows the same texts as y versus x . The primary evaluation therefore contains $3,750 \times 2 = 7,500$ physical judgments. Each response contains an A/B/tie verdict, confidence on a 1–5 scale, and a short comparison-specific justification. Confidence is recorded for diagnostics but never used to weight primary utility. The swapped design does not make the two judgments statistically independent; its purpose is to make presentation-order instability observable. For execution, each campaign/pass bundle is evaluated in a fresh Temporary Chat, with multiple comparisons processed within that context. Pass A and Pass B are evaluated in separate fresh chats. The 7,500 judgments should therefore be understood as dense repeated measurement over 50 experimental tasks, not as 7,500 independent samples.

5.4 From Pairwise Judgments to Candidate-Set Utility

For one physical presentation, a verdict is mapped back to the identity of candidate x :

$$v(x, y) = \begin{cases} 1, & x \text{ preferred,} \\ 0.5, & \text{tie,} \\ 0, & y \text{ preferred.} \end{cases}$$

Let $v_1(x, y)$ and $v_2(x, y)$ denote the two exact swapped presentations after both are mapped back to the same candidate identity. Their average defines the canonical logical-pair score

$$s(x, y) = \frac{v_1(x, y) + v_2(x, y)}{2},$$

so that

$$s(x, y) \in \{0, 0.25, 0.5, 0.75, 1\}.$$

A value of 1 means both orientations prefer x ; 0 means both prefer y . A value of 0.5 can arise either from two ties or from compensating orientation-sensitive outcomes. The intermediate 0.25 and 0.75 values occur when a tie in one orientation accompanies a directional preference in the other.

For an AIRTIST candidate a_i evaluated against a five-candidate control pool B ,

$$U_B(a_i) = \frac{1}{5} \sum_{b_j \in B} s(a_i, b_j).$$

This quantity is the candidate’s external utility against that opponent pool. It is external in the operational sense that AIRTIST’s search and ranking procedures never observe it.

The corresponding complete-pool preference for one brief is

$$P(A5 > B) = \frac{1}{25} \sum_{i=1}^5 \sum_{j=1}^5 s(a_i, b_j),$$

which is equivalently the mean external utility of the five A5 candidates against that control pool.

For interpretability we also report a centered representation,

$$\Delta U = 2P(A5 > B) - 1.$$

Here, $\Delta U = 0$ denotes parity between the two five-proposal populations, $+1$ would mean that every swapped comparison favors A5 in both orientations, and -1 the reverse. Preference and centered effect are not two independent metrics. They are two scales for the same complete 5×5 comparison matrix.

5.5 Measures for the Three Research Questions

The metric hierarchy is deliberately explicit. We distinguish the primary outcome of each experiment from secondary diagnostics that characterize the same underlying candidate set or ranking. A secondary statistic becoming significant does not retroactively turn it into an additional primary hypothesis.

RQ1: sampled-set quality The primary RQ1 outcome is the brief-level mean complete-pool effect ΔU defined above. It is computed separately for A5 vs. B5raw, A5 vs. B5self, A5 vs. M5. The first two comparisons measure improvement relative to simpler stages of the same generation process; the third is the central shared-parent mechanism comparison.

Mean pool quality alone, however, does not tell us how improvement is distributed among the five proposals. A system could increase its average because one exceptional candidate becomes much stronger while the remaining proposals stagnate or deteriorate. We therefore report four additional oracle distributional diagnostics. For each pool, candidate utilities are computed against the opposing five-candidate pool. We then identify:

- Best-candidate utility: the utility of the externally strongest candidate in the five-proposal set.
- Top-three mean utility: the average external utility of the three strongest candidates.
- Worst-candidate utility: the utility of the externally weakest candidate.
- Top-heaviness: the gap between the externally best candidate’s utility and the mean utility of its own pool.

Differences between A5 and each control on these quantities describe whether the pool shift is concentrated at the top, extends through several candidates, or reaches the lower tail.

The word oracle is important here. The identities of the externally best, top-three, and worst candidates are determined after judging from external utility. They are not AIRTIST’s internal ranks and do not measure whether the system knew which candidate was best. That question belongs to RQ3. These distributional quantities are secondary diagnostics of RQ1. They are not additional members of the primary three-contrast hypothesis family.

RQ2: competitive intervention RQ2 separates what the refinement policy does from whether its adopted changes improve over what they replace. The first quantity is descriptive. Across all five lineages for every brief, we record whether the final proposal state is content-identical to the B5-self incumbent, a selected focused reconstruction, or a selected structural escape. An unchanged final state should not be interpreted as a semantic “Keep” decision in every case: it can also occur because no challenger remains eligible after gating. We therefore describe this statistic conservatively as the proportion of final lane states that remain identical to the incumbent. The stronger RQ2 test concerns elected replacements. Because A5 and B5-self share exact parent identities and the A5–B5-self evaluation is exhaustive, every selected challenger already has a blind external comparison against the specific incumbent proposal from which it was generated.

For a selected replacement c and its exact parent p , the parent-relative score is $s(c, p)$, and its centered form is

$$\delta(c, p) = 2s(c, p) - 1.$$

A positive value favors the selected challenger; a negative value favors the displaced parent. Because some briefs contain several selected replacements while others contain none, the primary inferential quantity is brief-balanced and conditional on intervention. We first average the centered exact-parent effects of all selected replacements within a brief. Briefs in which no lineage is replaced do not contribute to this conditional replacement-quality estimand; their behavior is represented instead by the separate retention statistic. Statistical inference is then performed across the resulting brief-level replacement effects.

We additionally report raw replacement preference and role-specific outcomes for focused and structural replacements. These role-specific figures are descriptive. Focused and structural challengers are generated and selected under different local circumstances, so a higher observed preference for one role does not by itself establish a causal advantage of that operator. RQ2 therefore supports a deliberately bounded inference: it can test whether interventions actually adopted by the frozen policy tend to beat their exact parents. It does not compare AIRTIST with a counterfactual policy forced to replace every incumbent, because that counterfactual candidate set was not generated.

RQ3: terminal selection on an identical candidate set RQ3 removes generation as a source of variation entirely. For every brief, the Base selector and the Hybrid tournament selector rank exactly the same five frozen A5 proposal texts. Neither selector observes the external Sol judgments used for evaluation. The Base selector is the frozen content-only terminal ranker used in the ablation. It receives the brief, constraints, and five authored dossiers, scores each proposal on the same ten equal-weight internal quality dimensions, applies the narrow terminal hard-veto tier, and orders admissible candidates by their mean pointwise score. The bounded top-two comparator is disabled in this condition. The Hybrid selector is the terminal policy described in Section 4. It preserves the same content-only information boundary but adds an independent pointwise stage followed by pairwise comparison among up to the three highest-scoring non-veto finalists. Finalist pairs are judged under exact A/B reversal and aggregated into a tournament ordering before the complete five-item order is returned. Thus RQ3 changes the ranking policy, not the candidate texts available to it.

To score either selector, each A5 candidate is assigned an aggregate external utility over all 15 baseline opponents:

$$U(a_i) = \frac{1}{15} \sum_{b \in B5_{\text{raw}} \cup B5_{\text{self}} \cup M5} s(a_i, b).$$

Thus the external relevance signal used for RQ3 incorporates all three complete cross-pool matrices while remaining unavailable to the selectors. Let $\pi = (\pi_1, \dots, \pi_5)$ denote a selector’s ordering of the five A5 candidates. The primary RQ3 outcome is

$$\text{Utility@1}(\pi) = U(a_{\pi_1}),$$

the external utility of the proposal the selector places first. This metric directly captures the attention-allocation problem motivating terminal ranking: given the exact same proposal set, does one selector surface an externally stronger first option?

We additionally report several secondary ranking diagnostics. Regret@1 measures the exter-

nal utility sacrificed relative to an oracle top choice:

$$\text{Regret@1}(\pi) = \max_i U(a_i) - U(a_{\pi_1}).$$

For a fixed candidate pool, Regret@1 is algebraically coupled to Utility@1; it is useful because it expresses the top-ranking error as lost utility, but it is not an independent confirmation of a Utility@1 effect. Exact top-1 accuracy asks whether the proposal ranked first attains the maximum external utility in its pool. If several candidates share the maximum, selecting any of them counts as correct. Hit@2 and Hit@3 ask whether at least one externally best candidate occurs within the first two or first three positions. NDCG@5 treats external utility as graded relevance and evaluates the quality of the complete five-candidate ordering, giving greater weight to errors near the top. Utility@1 is the primary selector endpoint. Regret, top-k coverage, and NDCG characterize different practical aspects of the ranking but are not treated as a collection from which whichever metric happens to reach significance can substitute for the primary outcome.

Table 13: Metric hierarchy. Post-hoc semantic analyses are intentionally excluded from the primary metric set and are introduced where used in Results.

Analysis	Primary inferential quantity	Secondary / descriptive quantities
RQ1	Mean complete-pool ΔU for three planned control contrasts	Best, top-three, worst, top-heaviness; brief W/T/L
RQ2	Brief-balanced selected-replacement effect vs. exact parent	Incumbent-identical / focused / structural frequencies; pair preference; role-specific outcomes
RQ3	Paired difference in Utility@1 on identical A5 pools	Regret@1, Top-1, Hit@2, Hit@3, NDCG@5
Selected-output check	Holistic selected-pair preference	Directional consistency with complete-pool results
Measurement robustness	No new primary endpoint	Swap stability, position, consensus-only, verbosity, and leave-out sensitivity

Analyses introduced after inspection of the frozen experiment—most notably latent-semantic pool breadth, the relationship between quality gain and breadth, and parent-relative semantic distance between focused and structural challengers—are not part of this primary metric set. They are introduced explicitly as post-hoc diagnostics at their first use in the Results and Mechanism Analysis sections.

5.6 Statistical Inference

The brief is the statistical unit throughout the primary analysis. This point is important because exhaustive comparison produces many judgments from each task. For RQ1, one brief contributes 25 logical candidate comparisons for each control, and each logical comparison is observed in two presentation orientations. These observations provide a denser estimate of the candidate sets associated with that brief; they do not convert one artistic problem into dozens of independent benchmark tasks. Accordingly, pair-level scores are first aggregated to the relevant brief-level statistic before uncertainty is estimated.

Confidence intervals use 10,000 brief-level bootstrap resamples. Resampling is performed over briefs rather than over individual candidate pairs. For paired hypotheses, we use a paired randomization test at the brief level: under the null of no systematic directional effect, the sign of each brief-level paired difference is randomly reversed. The resulting reference distribution is used for two-sided significance testing.

The three planned RQ1 complete-pool comparisons $A5 - B5_{\text{raw}}$, $A5 - B5_{\text{self}}$, $A5 - M5$ form one hypothesis family and are corrected using Holm’s procedure. No larger omnibus correction is retroactively constructed across all quantities in the paper. RQ2 exact-parent improvement and RQ3 Utility@1 are distinct experiments with their own primary estimands. Distributional pool diagnostics, secondary ranking metrics, and robustness analyses retain their explicitly secondary or diagnostic status rather than being used as alternative routes to a positive primary claim.

For interpretability, we also report brief-level win/tie/loss counts. These are based on the sign of the corresponding brief-level effect and make heterogeneity visible even when the aggregate mean is positive. Judge-reported confidence never changes the primary score, the bootstrap weight, or the statistical unit.

5.7 Secondary Selected-Output Evaluation

Complete-pool evaluation deliberately removes the system’s internal ranker from RQ1. It therefore answers a question about the quality of the available five-proposal set, not the more familiar end-to-end question of what happens if a user attends only to the proposal surfaced first. We examine that question in a separate blind selected-output campaign. For each held-out brief, we take the proposal ranked first by AIrtist’s frozen Hybrid terminal selector and compare it separately with the top-ranked proposal from each of the three control conditions under their frozen ranking artifacts: A5top vs. B5self,top, A5top vs. M5top, A5top vs. B5raw,top. This yields three logical selected-output comparisons per brief, or 150 logical comparisons across 50 briefs. Each is again evaluated under exact A/B reversal, yielding 300 physical judgments.

The selected-output campaign is run separately from the complete-pool evaluation but uses the same blind holistic judging criterion and commission context. The external judge sees the brief, constraints, and two complete dossiers, but not system identities, source pools, candidate IDs, internal ranks, or generation histories.

This experiment is deliberately secondary. For one system contrast, complete-pool evaluation observes 25 cross-pool relationships per brief; the selected-output check observes only one. It consequently discards most of the information available about the sampled candidate set and reintroduces dependence on each system’s internal ranking. Its value is instead interpretive. If the complete-pool advantage and the selected-output comparison point in the same direction, the former has not obviously been created by improvements that disappear as soon as the system is collapsed back to the single-output presentation a user is most likely to notice first.

5.8 Measurement-Robustness Analyses

The primary evaluation relies on a language-model judge, so the experiment separately examines several identifiable ways in which that measurement could mislead.

First, the exact-swapped design makes presentation-order stability observable. We report agreement between the two canonical A/B orientations as well as the aggregate preference for whichever proposal happened to be physically displayed as Candidate A. The latter should be near 0.5 in the absence of a gross global position preference.

Second, for the central A5–M5 comparison we perform a consensus-only sensitivity analysis, restricting the matrix to logical pairs for which the two exact orientations yield the same canonical verdict. This is not a cleaner replacement estimator: agreement filtering changes the population of comparisons and may preferentially retain easier cases. Its purpose is narrower—testing whether the central effect depends on the comparisons whose verdict visibly reverses under presentation order.

Third, we examine relative verbosity. Structured agentic pipelines may produce longer dossiers, and model judges can reward elaboration independently of substance. We therefore measure candidate length, test its association with pairwise preference, and examine the A5–M5 direction in comparisons where A5 is no longer than its opponent and where it is strictly shorter. These subsets are robustness diagnostics rather than additional confirmatory endpoints.

Fourth, we test dependence on benchmark composition through leave-one-brief-out and leave-one-category-out re-estimation of the central matched-control effect. These analyses ask whether the overall direction is explained by one unusually favorable task or one complete coverage category; they are not used to infer that AIrtist performs differently across artistic categories.

6 Results

6.1 RQ1: Domain-Structured Development Improves the Sampled Possibility Set

Table 14 reports the three planned complete-pool contrasts. For every brief and control condition, all five A5 proposals are compared with all five control proposals. Mean complete-pool preference therefore reflects the entire 5×5 relation between two sampled sets rather than either system’s selected top output.

All three planned contrasts favor A5. The largest effect is against raw generation. A single generic self-refinement pass narrows the difference but does not remove it. Most importantly, A5 also exceeds M5, the substantially matched control that begins from the same five B5-self parent texts, maintains the same five lineages and two challengers per lineage, leaves the incumbent eligible, and shares the challenger-gating and lane-comparison procedures.

The A5–M5 comparison is therefore the central RQ1 result. Identical parents rule out a more

favorable starting sample as the explanation. Matching much of the iterative scaffold also makes a simple “more continuations from the same parents” account substantially less plausible. The principal designed differences are A1rtist’s domain-structured evidence, opportunity-state representation, preservation constraints, and differentiated focused and structural search roles. The residual +0.0836 effect consequently supports a package-level contribution from domain-structured local search relative to substantially matched generic iteration. It does not isolate the causal contribution of any individual component, nor does it estimate improvement per token or model call.

The aggregate result is heterogeneous rather than universal. Against M5, A5 has a positive brief-level effect on 31 of 50 briefs, ties on 8, and a negative effect on 11. Figure 2 exposes this distribution rather than reducing it to its mean. The leave-one-brief analysis reported below provides the stronger robustness statement: no single brief determines the sign of the matched-control estimate.

Table 14: Primary complete-pool results over 50 held-out briefs. All three planned contrasts remain significant after Holm correction.

Comparison	A5 pref.	ΔU	95% CI	Brief W/T/L	Holm p
A5 vs B5-raw	0.6068	+0.2136	[+0.1412,+0.2848]	42/0/8	< .0001
A5 vs B5-self	0.5502	+0.1004	[+0.0488,+0.1512]	32/11/7	.0008
A5 vs matched M5	0.5418	+0.0836	[+0.0340,+0.1304]	31/8/11	.0015

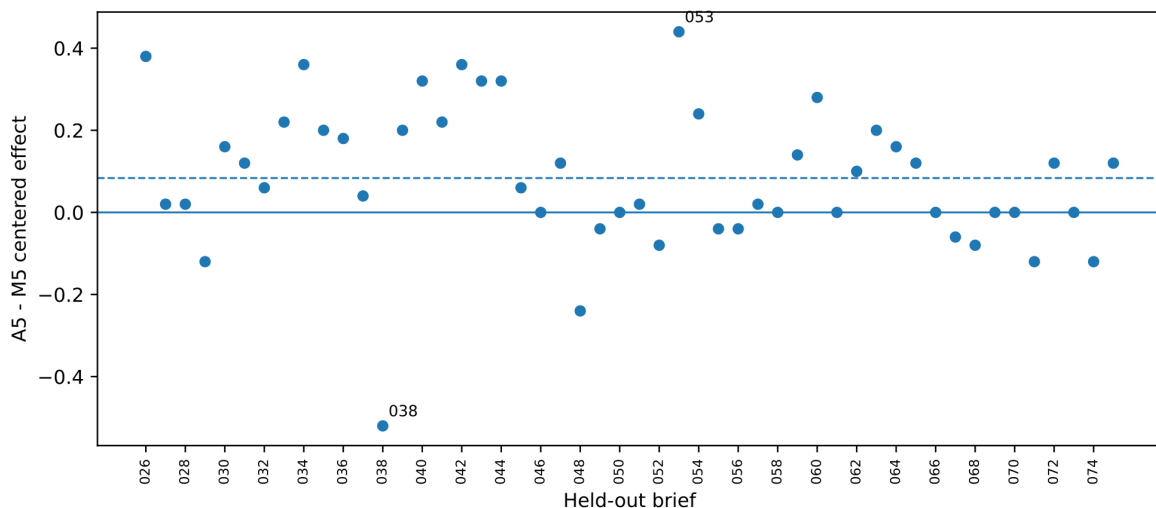


Figure 2: Matched-control effect across all 50 held-out briefs. Each point is one brief-level centered A5-M5 complete-pool effect; the dashed horizontal line is the overall mean. The aggregate effect is a distributional shift, not dominance on every commission problem. Briefs 038 and 053 are annotated because they return later as qualitative mechanism cases.

6.1.1 The gain is not confined to an oracle-best candidate

A higher five-proposal mean can still arise in different ways. Structured search might produce one exceptional proposal while leaving the rest of the pool unchanged or weaker, or it might shift a larger portion of the sampled set. Section 5 therefore defined the externally best candidate, top-three mean, externally worst candidate, and top-heaviness as secondary oracle diagnostics.

The matched M5 comparison is particularly informative. The oracle-best difference is positive but imprecisely estimated (+0.0560, 95% CI [-0.0020, +0.1150]). In contrast, the top-three mean differs by +0.0767 (unadjusted 95% CI [+0.0173, +0.1363]), and the externally worst proposal differs by +0.0710 ([+0.0250, +0.1180]). Top-heaviness does not increase (-0.0276, [-0.0670, +0.0114]).

These diagnostics are secondary and their intervals are not members of the Holm-corrected primary family. Nevertheless, their pattern is informative. The matched-control advantage is not confined to the oracle maximum: it is visible in the pool mean, the upper three candidates, and the lower tail, while there is no detected increase in top-heaviness. The pattern is therefore inconsistent with a purely top-heavy account in which the aggregate gain is carried only by an isolated lucky maximum.

Table 15: Distributional diagnostics for RQ1. These are secondary oracle descriptors of the externally evaluated pools, not selector ranks.

Control	Best Δ	Mean Δ	Top-3 Δ	Worst Δ	Top-heaviness Δ
B5-self	+0.0660	+0.1004	+0.1190	+0.0530	-0.0344
M5	+0.0560	+0.0836	+0.0767	+0.0710	-0.0276
B5-raw	+0.2060	+0.2136	+0.2123	+0.1750	-0.0076

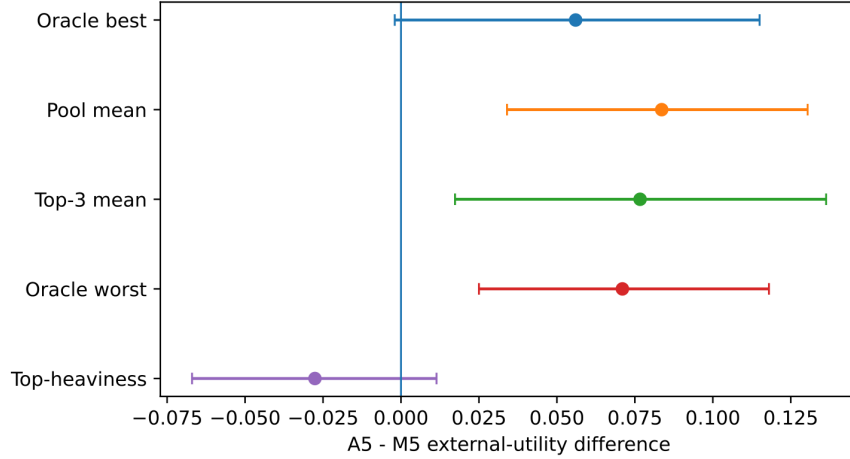


Figure 3: Where the A5-M5 difference occurs within the sampled possibility set. Points show brief-balanced differences with 95% brief-bootstrap intervals. Pool mean is the planned primary endpoint; the remaining quantities are secondary oracle diagnostics with unadjusted intervals.

6.1.2 Post-hoc: higher measured quality does not imply greater latent-semantic breadth

Section 2 deliberately separated sampled-set quality from breadth. Maintaining five lineages does not establish that they occupy a wider proposal space, and higher external preference does not logically imply greater diversity. We therefore examine this distinction post hoc.

Using the primary semantic representation—artistic operation, spatial logic, material logic, and spectator encounter—A5 pools have lower mean within-pool latent-semantic distance than M5 pools. The paired A5-M5 breadth difference is -0.0554 (95% CI [-0.0744, -0.0385]); A5 is more concentrated on 43 of 50 briefs. Adding concept description to the representation preserves the qualitative direction.

This finding should not be generalized to “diversity” without qualification. It concerns one explicitly defined latent-semantic representation. Under that representation, however, higher measured proposal quality does not imply greater breadth. At the brief level, Pearson correlation between quality gain and breadth difference is $r = -0.229$ ($p = .110$), while the exploratory rank association is Spearman $\rho = -0.331$ ($p = .019$). Twenty-nine briefs combine a positive A5-M5 quality effect with lower A5 breadth, whereas only two combine positive quality effect with greater breadth. We make no causal claim from this post-hoc association: it does not show that semantic concentration causes quality improvement or that wider search would be undesirable.

6.2 RQ2: Selected Interventions Tend to Improve on Their Exact Parents

Across 250 lineages, 149 final states remain content-identical to their incumbents (59.6%), 66 end in focused reconstructions (26.4%), and 35 in structural escapes (14.0%). Thus 101 lineages end in a new proposal state. The 59.6% figure is a property of the complete refinement pipeline rather than a count of explicit semantic “Keep” decisions: a final state can remain identical because challengers fail eligibility or because the incumbent survives subsequent competition.

Because A5 and B5-self retain exact lineage identity, every adopted challenger has an existing blind comparison with the specific parent text it displaced. Across 101 selected replacements, external preference for the challenger is 0.693. After averaging multiple replacement effects within each brief, the brief-balanced centered effect is +0.437 (95% CI [+0.266, +0.594], paired randomization $p < .0001$); brief W/T/L is 31/8/7.

Among 73 selected-replacement comparisons whose exact A/B orientations return the same verdict, challengers beat their parents 56 times and lose 17 times. The remaining 28 replacement

comparisons are orientation-sensitive. The RQ2 result is therefore stronger than the descriptive retention frequency: conditional on the frozen policy adopting a replacement, selected challengers are externally preferred on average to the exact states they replace. The experiment does not contain a mandatory-replacement counterfactual and therefore does not establish that incumbent retention itself causes the gain.

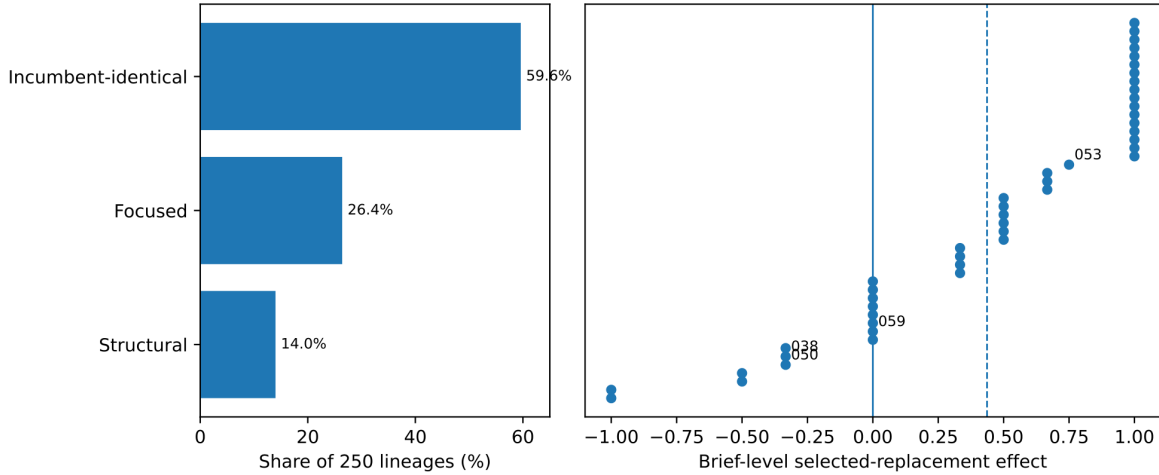


Figure 4: Competitive refinement separates whether a lineage changes from whether the adopted change helps. Left: final states over 250 lineages. Right: brief-level exact-parent effects among briefs with at least one selected replacement, sorted by effect; the dashed line is the brief-balanced mean. Labels mark cases used later in the mechanism analysis.

6.2.1 Focused and structural roles produce different transformation magnitudes

The selected focused and structural replacements have parent-relative external preferences of 0.659 and 0.757, respectively. These values are descriptive because the two roles are generated and selected under different local conditions.

A more direct post-hoc question is whether the two prompts produce different magnitudes of movement from the same exact parents. Under the primary semantic representation, mean parent-to-Focused distance is 0.1834 and parent-to-Structural distance is 0.2409. The paired Structural-minus-Focused difference is +0.0574 (95% CI [+0.0429, +0.0734]); Structural outputs are farther from their parents on 45 of 50 briefs. This is consistent with the transformation permissions defined in Section 4. It does not establish that larger moves are artistically better or that the prompts instantiate formally disjoint operators.

6.3 RQ3: Selection Leaves Substantial Recognition Headroom; the Hybrid Advantage Is Uncertain

Two findings should be distinguished. First, the frozen pools contain a sizeable measured recognition gap. Averaged over briefs, the externally highest-utility candidate already present within the five-proposal A5 set has utility 0.8660. Hybrid surfaces proposals with mean Utility@1 0.6417, leaving mean Regret@1 0.2243. It selects an externally highest-utility candidate first on 13 of 50 briefs (26%), and on 15 briefs no externally highest-utility candidate appears even in its top three. These values concern an external model-based diagnostic, not an objective claim about which artwork an artist should choose.

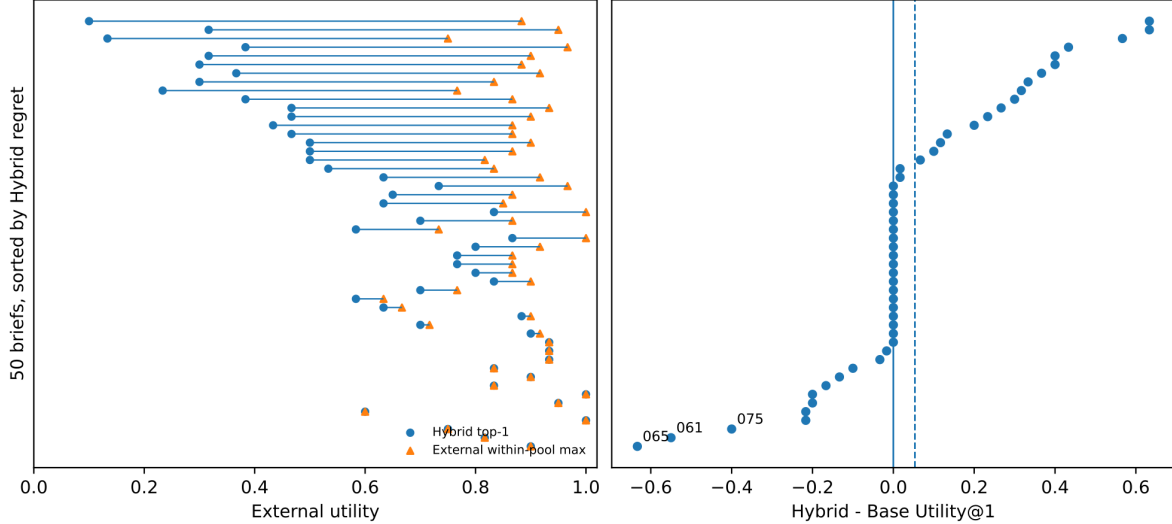
Second, the experiment does not establish that Hybrid reliably solves this problem better than Base. Hybrid has favorable point estimates on Utility@1, regret, exact top-1, top-2 coverage, and NDCG. Its primary Utility@1 advantage is +0.0533, however, with 95% CI [-0.0160, +0.1240] and paired randomization $p = .151$. Hybrid yields higher Utility@1 on 19 briefs, the same utility on 19, and lower utility on 12. The appropriate RQ3 conclusion is therefore not selector superiority: selection is empirically separable from search and remains imperfect, while the present tournament policy shows a favorable but statistically unresolved advantage over the simpler Base selector.

6.4 Secondary Selected-Output Check

Complete-pool evaluation is the primary search endpoint because it observes all 25 cross-pool relationships for each brief. A separate blind campaign asks a more conventional end-

Table 16: Terminal-selection metrics on identical frozen A5 proposal sets.

Metric	Base	Hybrid
Utility@1 $\uparrow$	0.5883	0.6417
Regret@1 $\downarrow$	0.2777	0.2243
Exact top-1 accuracy $\uparrow$	0.18	0.26
External maximum in top-2 $\uparrow$	0.44	0.48
External maximum in top-3 $\uparrow$	0.70	0.70
NDCG@5 $\uparrow$	0.8758	0.8954

**Figure 5:** Two views of terminal selection on identical frozen A5 pools. Left: Hybrid top-1 utility and the external within-pool maximum, with briefs sorted by Hybrid regret; horizontal segments show the measured headroom already present in the generated set. Right: Hybrid-minus-Base Utility@1 by brief; the dashed line is the mean +0.0533 difference, which is not statistically resolved. Briefs 061, 065, and 075 are the three largest regressions and are examined in Section 7.3.

to-end question: if each condition is collapsed to the single proposal it surfaces first, which selected proposal receives higher holistic preference? Across the 50 held-out briefs, pooled point estimates favor Hybrid-selected A5 in all three contrasts (Table 17).

Table 17: Secondary selected-output holistic comparison over 50 held-out briefs. Values are pooled descriptive preference point estimates for Hybrid-selected A5. Brief-level uncertainty is not reconstructed because the complete per-brief records required for that analysis are unavailable for part of this secondary campaign.

Selected-output comparison	A5 preference
A5 vs. selected B5-self	0.630
A5 vs. selected M5	0.620
A5 vs. selected B5-raw	0.670

We treat these values as a descriptive secondary consistency check and make no separate inferential claim from them here. We therefore retain this campaign as a descriptive directional check and do not reconstruct brief-level uncertainty from incomplete per-brief records. Each contrast observes only one selected pair per brief and reintroduces dependence on internal ranking; the exhaustive pool analysis therefore remains the stronger evidence for RQ1.

6.5 Robustness of the External Model-Based Measurement

The preceding effects are properties of a model-mediated holistic evaluation. The analyses below ask whether the central A5–M5 result is readily explained by several observable features of that measurement process. They do not establish equivalence to professional artistic or curatorial judgment.

Across all three complete-pool campaigns, exact A/B orientations agree on 79.41% of the 3,750 logical candidate pairs. When both orientations report confidence 4–5, agreement rises to 94.73% across 1,803 pairs. The global preference for whichever proposal happens to occupy physical

Candidate-A position is 0.5016, close to positional parity. These diagnostics make a simple global position-bias account difficult to sustain, while leaving subtler evaluator biases possible.

For the central A5–M5 comparison, 962 of 1,250 logical pairs are swap-consistent. Restricting the analysis to this different, more stable subset yields A5 preference 0.5608 and a brief-balanced centered effect of +0.1280 (95% CI [+0.0639, +0.1916], randomization $p = .00024$; brief W/T/L 31/10/9). Consensus filtering changes the estimand and can preferentially retain easier comparisons, so this does not replace the full-matrix result. Its narrower implication is that the positive matched-control direction does not depend on comparisons whose verdict reverses under exact presentation inversion.

A5 dossiers are also longer on average than M5 dossiers (469.3 vs 425.1 words). Yet relative length is only weakly associated with pairwise outcome (Spearman $\rho = 0.045$, brief-cluster 95% CI [-0.040, +0.129]), and A5 preference remains above 0.5 when A5 is no longer than its M5 opponent (0.548) and when it is strictly shorter (0.555). These diagnostics make a simple monotonic preference for longer dossiers difficult to sustain without ruling out subtler stylistic effects.

Finally, removing each held-out brief in turn leaves the A5–M5 point estimate positive, ranging from +0.0763 to +0.0959. Removing any complete ten-brief coverage category also preserves the positive direction, with re-estimated effects from +0.0740 to +0.0955. These are sensitivity ranges rather than confidence intervals or category-level hypothesis tests.

Taken together, these checks make several narrow procedural explanations difficult to sustain: a global physical-position preference, dependence on visibly order-unstable pairs, one influential brief, one coverage category, or a simple monotonic preference for longer dossiers. These robustness checks do not establish construct or ecological validity; see Sections 10.1–10.2.

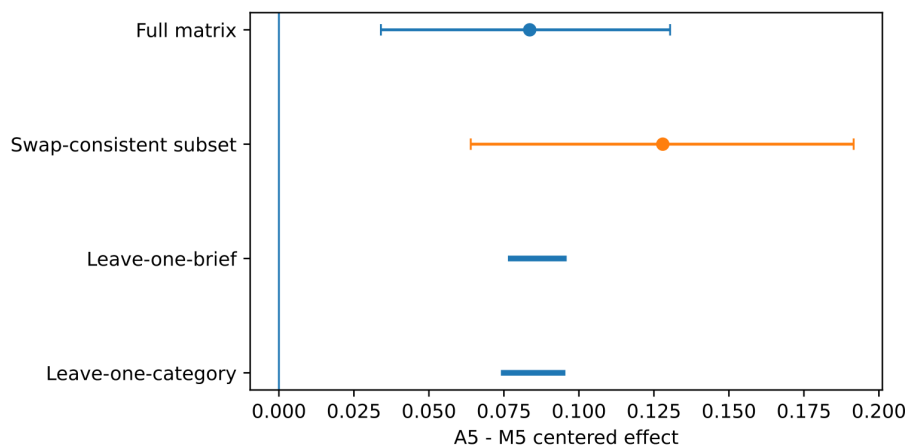


Figure 6: Sensitivity of the central A5-M5 effect. Full and swap-consistent rows show point estimates with 95% bootstrap intervals. Leave-one-brief and leave-one-category rows show only the range of re-estimated point effects after omission; they are not confidence intervals.

7 Mechanism and Failure Analysis

Quantitative effects do not by themselves show what selected transformations change or where failures enter the pipeline. We therefore inspect frozen lineage and ranking traces selected for explanatory contrast rather than statistical representativeness; they add no independent observations to RQ1–RQ3.

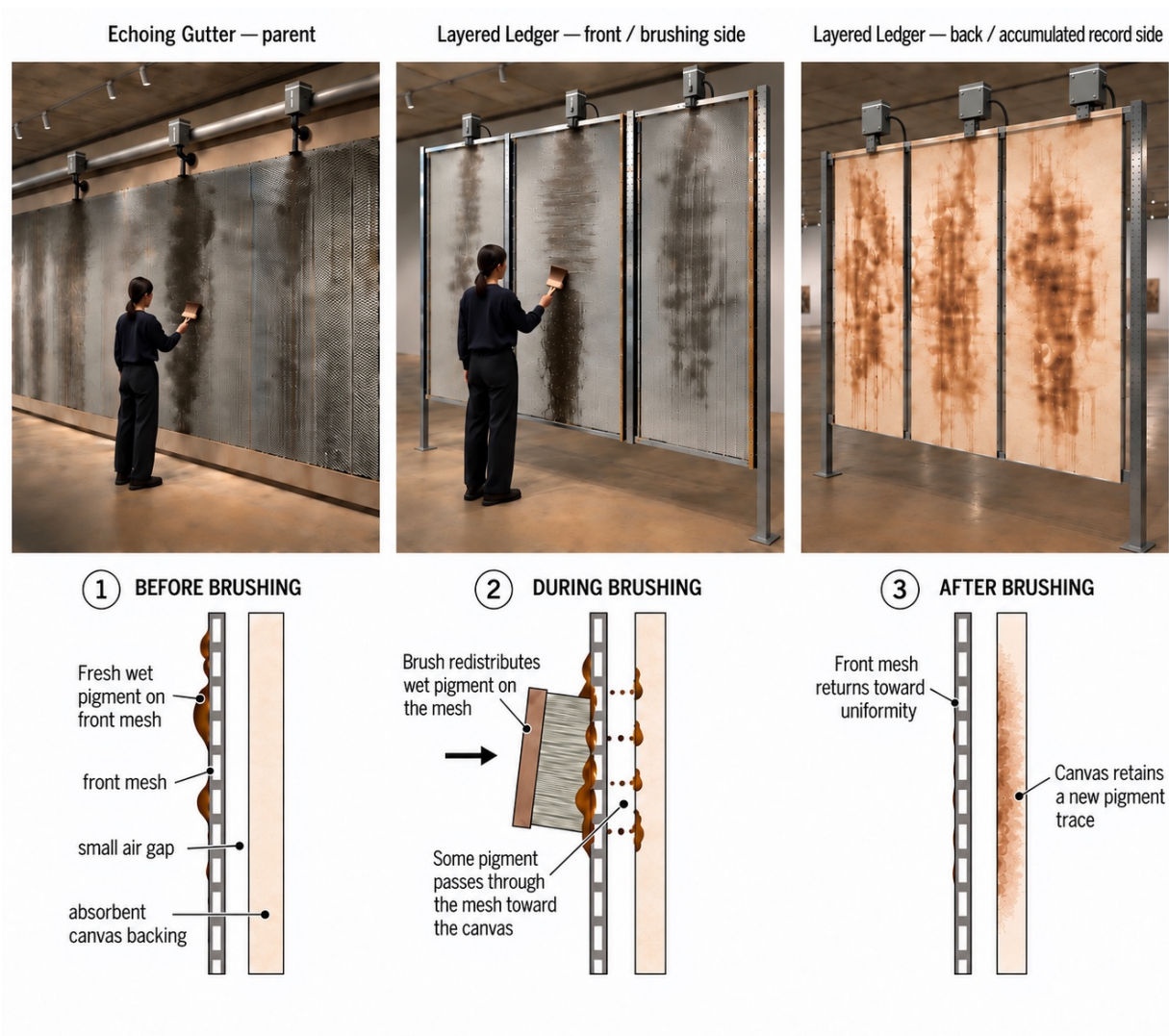


Figure 7: Post-hoc explanatory visualization of Brief 053. Top: parent Echoing Gutter; selected Structural challenger Layered Ledger from the front/brushing side and the rear/accumulated-record side. Bottom: one plausible fabrication interpretation of the transfer mechanism: brushing redistributes the wet trace on the mesh while part of the pigment passes through the openings and is absorbed into the canvas, where it can remain visible from the reverse side. The mesh–canvas gap, rear access, and pigment-delivery hardware are illustrative production choices added after evaluation; the external evidence remains the blind exact-parent text comparison.

7.1 What Selected Transformations Change

Section 2 introduced the scaffold stakes →artistic operation →realization. The lineage traces make this distinction concrete. A useful intervention need not invent a new topic. It may instead change the relation through which the same pressure becomes operative, shift where causality resides, reorganize the spectator’s spatial relation, or make a production constraint constitutive rather than compensatory.

7.1.1 Brief 053: repair stops erasing its own history

Brief 053, introduced in Section 1, provides a particularly clear operation-level case. In Echoing Gutter, a reservoir releases pigmented water onto a six-meter galvanized-mesh curtain. A visitor then uses a broad brush to spread the wet streak back across the mesh, reducing the concentrated mark and returning the visible surface toward a more uniform state. The operation makes maintenance public, but it is largely reversible: the visible trace of each cycle is reduced rather than accumulated. Canvas, although required by the brief, remains peripheral.

Layered Ledger changes the material routing of the same maintenance act. Each panel places pigment-receptive canvas behind the galvanized mesh. While a newly released pigment line is still wet, brushing spreads the concentrated streak across the metal surface while pressing part

of the liquid through the mesh openings onto the canvas behind. Because the galvanized mesh is a relatively non-absorbent perforated surface while the canvas is absorbent, the same broad brush can thin and redistribute the wet layer on the mesh while forcing part of the pigment through the openings onto the backing. The front therefore returns toward visual uniformity, whereas the backing retains a new ghost mark. A fabricated version could tune mesh aperture, pigment viscosity and binder, canvas spacing, and the brush edge and pressure so that the front mark is redistributed while a controlled fraction reaches the backing; those are production parameters rather than changes to the artistic operation. One plausible fabrication would leave a small gap between mesh and canvas and tune the canvas–pigment system so that part of the transferred pigment wicks far enough through the canvas to remain visible from the reverse side. Allowing viewers to circulate around the panels would make that accumulated record directly legible; these are illustrative production choices rather than details evaluated in the frozen dossier. The operative difference is compact:

Parent	damage → public repair → visible return toward reset
Challenger	damage → public repair → front returns toward uniformity + history accumulated behind

The selected challenger receives an exact-parent score of 1.0: the external evaluator prefers it in both exact A/B orientations. The complete Brief-053 A5–M5 effect is +0.44. The latter is a property of all five proposals and is not evidence attributable to this one lineage; the case is used because the transformation itself is unusually easy to inspect.

7.1.2 Material causality: Brief 030

Brief 030 provides a different type of transformation. In *Chronicle of Pull*, each of twelve descending sandbag weights is mechanically coupled to a separate frozen analog clock whose gear train resists its descent; cables and pulleys transmit the resulting motion to different points of one shared voile curtain. Differences in gear resistance make the weights pull the curtain at different rates, so inherited obligation is made visible through a clock-and-weight apparatus.

The selected Structural challenger *Weight of Inheritance* removes both the clocks and the sandbags. Twelve reclaimed bricks of different masses instead pull different points of the same veil through steel cables and pulleys. The intended causal change is direct: unequal physical mass itself supplies unequal tension to the common surface. In fabrication, standard guides, bounded travel, and controlled braking or damping can be added as needed to turn those forces into slow, safe motion without changing that relation; the frozen proposal already specifies audience separation, redundant safety cables, pre-show checks, and manual release. Figure 8 shows one plausible fabrication interpretation of this relation; the specific cable routing, frame-mounted governing drums, redirect pulleys, and damping arrangement are explanatory production choices rather than details evaluated in the frozen proposal. The selected challenger wins both external orientations against its exact parent. The case is therefore a concrete example of material-causal reorganization, not evidence that larger transformations are generally better.

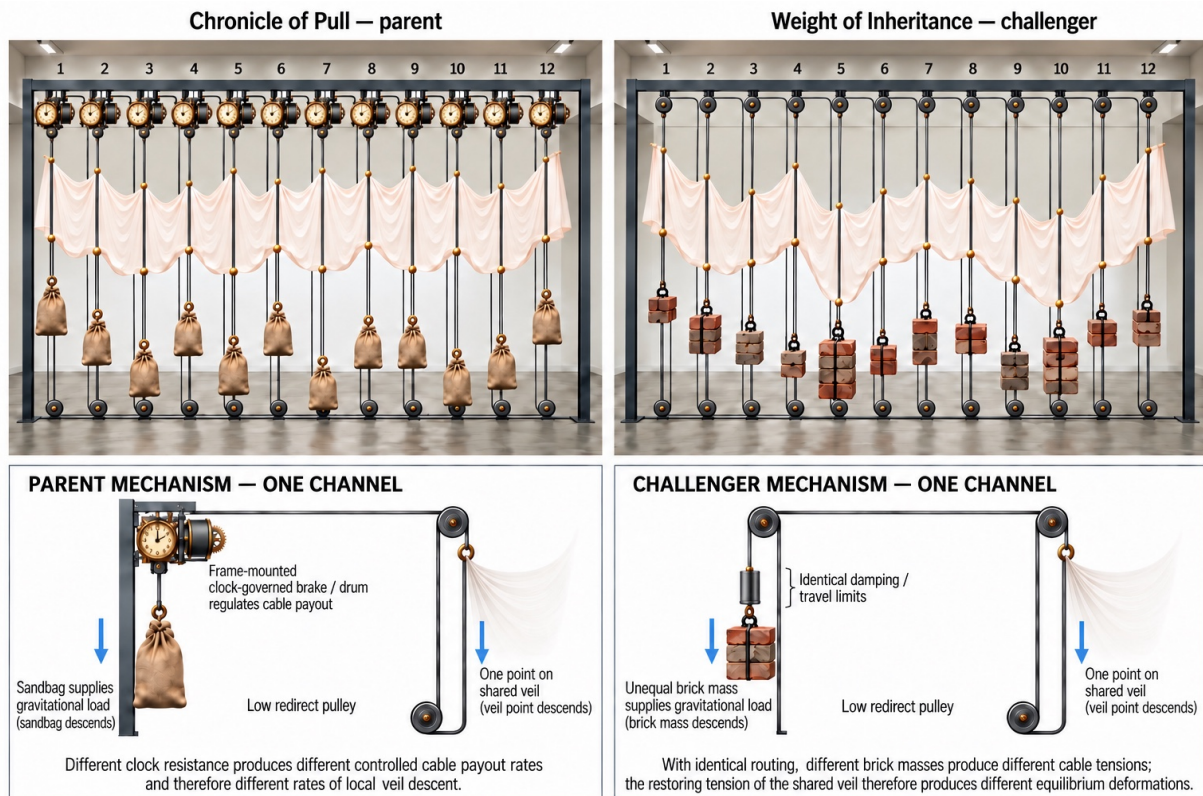


Figure 8: Post-hoc explanatory visualization of the selected Structural transformation in Brief 030, Chronicle of Pull → Weight of Inheritance. Top: matched installation views of the parent and challenger. Bottom: one plausible single-channel fabrication interpretation. In the parent, a sandbag supplies gravitational load while a clock-governed brake regulates cable payout; in the challenger, unequal reclaimed-brick masses supply unequal cable tension directly. Both act on the same shared veil through pulley routing. The specific brake/drum, redirect-pulley, damping, and travel-limit details are illustrative production choices added after evaluation; the external evidence remains the blind exact-parent text comparison.

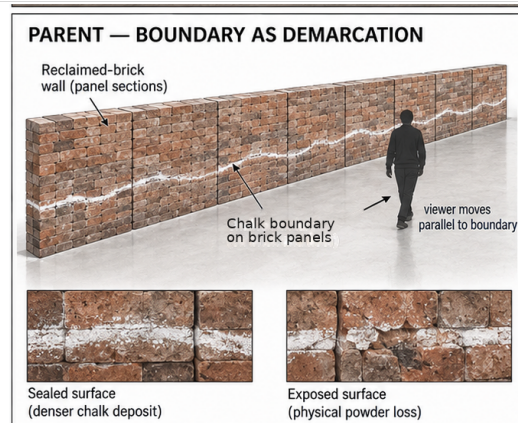
7.1.3 Spatial encounter: Brief 038

Brief 038 constrains the work to chalk, reclaimed brick, and rainwater-safe sealant and asks that a disputed boundary remain legible under partial weathering. In Eroding Chalk Line Labyrinth, reclaimed-brick panels line the gallery perimeter and a continuous chalk boundary snakes across their frontal surfaces. Selective sealant and periodic wetting are intended to make sections weather at different rates, while the visitor mainly walks parallel to the wall and watches the line fragment.

The selected Structural challenger Fragmented Borders: Erosion Field moves the same basic elements from the wall onto the floor. Low reclaimed-brick slabs form an irregular walkable field and the chalk boundary runs through it. The visitor must now cross, follow, or step around the line rather than observe it mainly from in front. A fabricated version would calibrate coating coverage or thickness, chalk application, and water delivery to produce staggered but legible weathering, while ordinary slab fixing, drainage, slip resistance, and an accessible route address floor safety. These production choices do not alter the central spatial change: boundary encountered as demarcation → boundary negotiated as terrain. The selected challenger wins both exact-parent orientations.

Yet the complete Brief-038 A5 set performs substantially worse than M5 ($\Delta U = -0.52$). This mismatch becomes useful in Section 7.2: a successful local intervention does not imply that the five-proposal search succeeded as a whole.

Eroding Chalk Line Labyrinth — parent



Fragmented Borders: Erosion Field — challenger

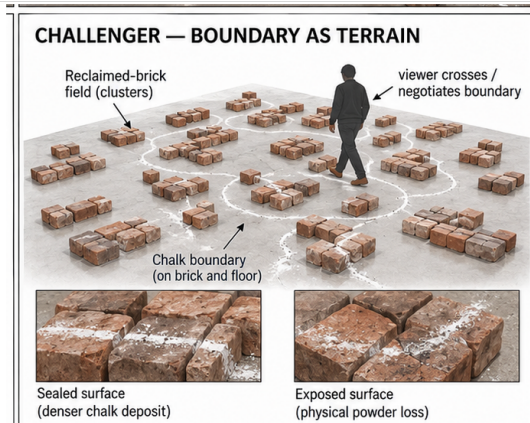


Figure 9: Post-hoc explanatory visualization of the selected Structural transformation in Brief 038, Eroding Chalk Line Labyrinth → Fragmented Borders: Erosion Field. Top: matched installation views of the parent and challenger. Bottom: simplified diagrams of the spatial and weathering logic. In the parent, a serpentine chalk boundary remains on reclaimed-brick perimeter panels and is encountered mainly while moving parallel to the wall; in the challenger, low reclaimed-brick slabs form a walkable field through which the chalk boundary passes, requiring visitors to cross and negotiate it. Differential chalk retention illustrates the proposed sealed/exposed weathering relation. The renderings and diagrammatic fabrication details were added after evaluation to clarify the textual proposals and provide no independent evidence of quality.

7.1.4 Institutional constraint as operative structure: Brief 059

Brief 059 asks for a university-commons installation about interrupted scholarship that can be paused for events, stored in labeled cases, and restored by non-specialist staff from a one-page diagram. In Retractable Lecture Shelf, a six-meter steel rail carries ten trays of printed scholarly texts. For an event, a motorized lift lowers the rail about 1.2 meters into a recessed floor cavity; afterward, staff raise it and restore the numbered tray order from a diagram. Such a mechanism is feasible where a suitable floor void or purpose-built recess is available, but the brief's explicit labeled-case storage requirement remains peripheral to the artistic action.

The selected Focused reconstruction, Fold-Out Fabric Archive, replaces the lift with ten fire-rated fabric panels on a lightweight hinged frame, each carrying a tray of essays. When displayed, the ten panels form a continuous vertical fabric wall along the corridor, with the essay trays visible across its surface. Staff release and fold the panels into a compact stack, place the folded elements in three labeled, color-coded cases in adjacent alcoves, and later follow a one-page diagram to unfold the structure and realign the trays. Hinge loads, tray retention, case packing, and fire/egress details are ordinary production-design questions rather than changes to the concept. The improvement is therefore not simply easier fabrication: pause → fold → case → restore becomes part of the work's own visible institutional behavior. The selected replacement wins both external orientations against its exact parent.

7.2 Contestable Refinement Can Succeed Locally, Fail Locally, or Leave the Parent Untouched

The strong mean RQ2 result can obscure an important fact: the 101 adopted interventions are not uniformly successful. Seventeen swap-consistent selected replacements lose to their exact parents. Brief 038 makes the resulting hierarchy of outcomes especially visible because

several different things happen inside the same five-lineage search. This brief is useful precisely because it resists a success-story reading. A coherent local improvement in one lineage neither guarantees that other interventions in the same brief are useful nor that the overall search policy constructs a stronger five-proposal set than another policy. RQ2 measures the consequence of a selected intervention relative to its exact parent; RQ1 evaluates the resulting candidate population as a whole.

Brief 050 supplies a cleaner single-lineage failure. Threaded Pause places a $3\text{ m} \times 2\text{ m}$ motorized loom in the gallery and invites volunteers to continue a shared tapestry. After five minutes, a safety-rated rotary cutter cuts the warp at its center and automatically stops the loom, leaving a visible rupture that ends that weaving cycle. A later cycle would require the loom to be re-threaded and reset; in fabrication the cutter can be isolated by an interlocked guard and operated as a scheduled supervised event. None of those ordinary safety and reset details changes the artistic event.

The opportunity state identifies a plausible problem: the timed cut may already read as a one-off gimmick. The selected Structural challenger Resonant Pause nevertheless keeps the same loom, the same five-minute wait, and the same cutter. At the instant of rupture it adds a sub-second burst of white noise from four speakers and a synchronized red LED flash. The proposed repair therefore leaves the diagnosed event intact and adds sensory emphasis around it rather than changing the underlying relation. The external exact-parent outcome is unambiguous: the selected challenger scores 0.0, meaning that the parent is preferred in both swapped orientations. A defensible diagnosis can still yield a proposed repair that intensifies rather than resolves the diagnosed weakness.

Table 18: Brief 038: local intervention outcomes within one five-lineage search. Exact-parent outcome is derived from the two swapped external comparisons.

Lane	Final state	Exact-parent outcome
1	Focused reconstruction	loss in both orientations
2	Structural escape	win in both orientations
3	Focused reconstruction	loss in both orientations
4	Incumbent retained	identity tie
5	Incumbent retained	identity tie
Complete five-proposal A5 vs M5 effect: -0.52		

7.3 Terminal Selection Errors Arise at Different Stages

The RQ3 aggregate identifies recognition headroom but does not reveal whether errors arise during broad pointwise screening or during the finalist tournament. A post-hoc stage-specific diagnostic first guards against cherry-picking individual tournament failures. Relative to the Hybrid pointwise top choice, the final tournament changes top-1 on 20 of 50 briefs. Its mean external Utility@1 change is +0.0150, with 95% bootstrap interval $[-0.0413, +0.0747]$; brief W/T/L is 9/30/11 and a paired randomization test is not significant ($p = .631$). Thus the experiment does not establish that tournament comparison is beneficial or harmful on average.

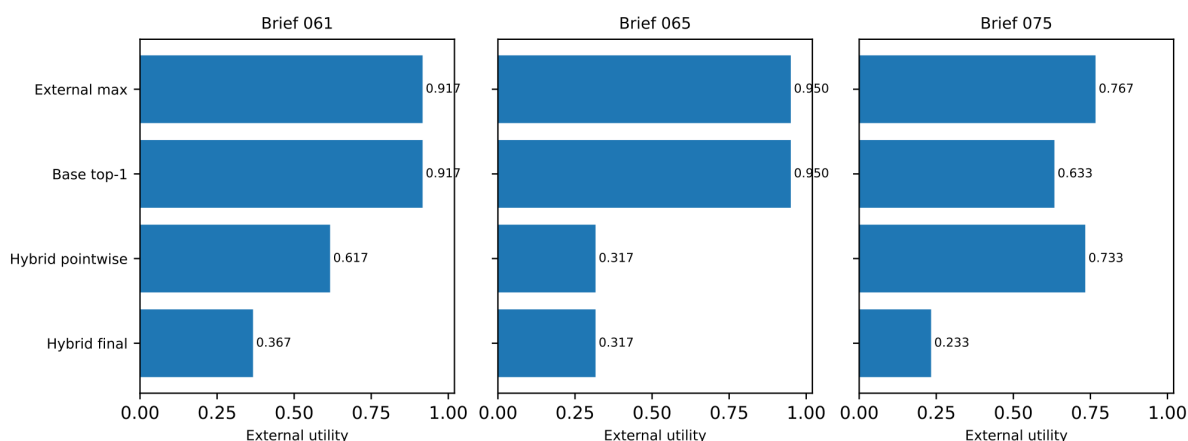


Figure 10: Three locations of terminal-selection error. Brief 061 loses the external maximum before tournament comparison; Brief 065 retains it among finalists but fails to recover it; Brief 075 regresses because tournament aggregation overturns a better pointwise choice. External utility is a diagnostic measured after generation, not an internal ranker score or a claim of objective artistic value.

Against that aggregate background, the three largest Hybrid-versus-Base regressions are informative because they localize three different failure paths:

- Brief 061 - screening failure. The externally highest-utility candidate, Weightless Ledger (0.917), is ranked outside the Hybrid finalist set. Base surfaces it first; Hybrid ultimately surfaces Morphing Transfer (0.367). The pairwise stage never gets the opportunity to recover the higher-utility candidate.
- Brief 065 - failure of recovery. Accumulating Ink Pool is the externally highest-utility candidate (0.950) and Base ranks it first. Hybrid pointwise ranking places it among the finalists but puts Midway Descent first; the tournament does not correct that ordering and the final top choice remains at 0.317.
- Brief 075 - tournament-induced regression. Hybrid pointwise ranking selects Echo Chamber (0.733), which would outperform the Base top proposal (0.633). Tournament aggregation overturns it and surfaces Heat Ledger (0.233); the external within-pool maximum is higher still at 0.767.

These cases localize distinct selection failures without implying that tournament comparison is harmful on average.

8 Discussion

8.1 What the Decomposition Reveals About Concept Development

For contemporary-art concept development, the central empirical result is not a single end-to-end score but the fact that three consequential stages behave differently under evaluation. A weak surfaced proposal may reflect a weak field of alternatives, an intervention that damaged a stronger state, or a failure to recognize which surviving proposal deserved attention. Once plausible proposals are inexpensive to generate, those failures are easily conflated by final-output evaluation.

The evidence is deliberately asymmetric: candidate construction and adopted replacements improve under the external measure, whereas terminal selection remains unresolved.

This asymmetry is useful beyond the domain because one compelling top-ranked output can conceal a weak candidate population or a harmful development policy, while a poor surfaced proposal can hide stronger material that already exists. Treating the sampled set, lineage transition, and terminal ranking as separate empirical objects therefore turns a property made especially visible by installation concept development into a more general diagnostic for creative agents.

The decomposition is operational rather than ontological. We do not claim that creativity consists of three independent modules, nor that these stages correspond literally to how artists develop work. Their value is experimental: they expose failure modes that an end-to-end winner-only evaluation would otherwise merge.

8.2 Search Semantics Matter in Addition to Search Width

The matched M5 comparison provides the strongest evidence that the organization of concept development matters beyond additional generic iteration. M5 starts from the same self-refined parents and preserves much of the same iterative scaffold, yet A5 retains a +0.0836 complete-pool effect. Computationally, this is evidence about local-search semantics: A5 converts critique into an opportunity state that distinguishes what should be preserved, what most needs intervention, which field attractor to escape, and which unused degree of freedom to exploit, then gives Focused and Structural challengers different permissions around that state.

The post-hoc operator analysis provides a behavioral consistency check on this interpretation. Structural candidates occupy farther parent-relative semantic neighborhoods than Focused candidates on 45 of 50 briefs under the chosen representation. This does not establish that operator heterogeneity causes the A5-M5 gain, nor that larger transformations are better. It does show that the two instructions are not merely different labels attached to indistinguishable rewrites: the intended difference in transformation scope is visible in their outputs.

Because M5 is substantially matched rather than compute-identical and several domain-structured components change together, the result supports a package-level effect rather than attribution to any individual mechanism (see Section 10.3).

More broadly, the result suggests a useful axis for inference-time search in open-ended domains. Search is often described through depth, branching factor, number of samples, critique rounds, or voting procedures. The present evidence indicates that the semantics assigned to a branch can matter alongside the amount of branching. Where revisions trade one virtue for another and no verified scalar reward is available, specifying what a move is allowed to change - and what it must protect - may be as consequential as allocating additional inference to the move itself. Whether this transfers beyond installation ideation remains an empirical question.

8.3 Additional Inference Can Test Whether to Change

Competitive refinement suggests a different use of additional inference: proposed revisions can be tested without making revision compulsory. Conditional on replacement, the frozen policy tends to adopt interventions preferred to the exact states they displace.

This distinction suggests a different role for additional inference in open-ended systems. Computation can be spent testing whether a current state deserves intervention, not merely searching for a successor that looks more developed. That design principle is potentially relevant wherever revision can exchange rather than monotonically accumulate virtues - for example in design, writing, planning, or research ideation - but the present experiment establishes it only for this proposal-development setting.

The experiment does not isolate the causal value of optional retention; doing so would require a matched mandatory-replacement policy (see Section 10.3).

8.4 Better Measured Candidate Sets Need Not Be Broader Candidate Sets

The post-hoc breadth analysis shows that multiplicity does not imply broader exploration: under the chosen latent-semantic representation, A5 is more concentrated than M5 despite its higher measured proposal quality.

This matters because sampled-set quality and breadth answer different questions. The quality analysis asks whether the retained proposals fare better under the external holistic evaluation. The breadth analysis asks how far apart proposals lie under one particular semantic representation. A system can improve one without improving the other. The present result therefore argues against using the existence of several agents, lanes, roles, or outputs as evidence that a system has preserved diversity.

The result should also not be inverted into a new optimization principle. It does not show that concentration causes higher quality, that narrower pools are preferable, or that A5 is less diverse in every meaningful artistic sense. The rank association between quality gain and breadth difference is exploratory, and latent-semantic distance captures only one notion of difference. Two proposals can be textually close while differing importantly in fabrication or institutional meaning; they can also be semantically distant while remaining variants of the same familiar artistic operation.

A more useful design implication is that quality and breadth may need separate controls. An open-ended creative system may want to retain several proposals above a quality threshold while explicitly preserving differences in operation, material strategy, or spectator relation, rather than assuming that a single search objective will maximize both. This also suggests a

stricter future evaluation: breadth should be measured directly and at several representational levels instead of inferred from branch count or agent multiplicity.

8.5 Selection Is Attention Allocation, and the Recognition Problem Remains Unresolved

The selector experiment establishes the recognition problem more clearly than the present solution: stronger measured material often already exists in the pool, while Hybrid’s advantage over Base remains statistically unresolved. This makes ranking an attention-allocation problem over already available material; Section 9.3 develops its artist-facing implications. Future work should examine ranking objectives, uncertainty-aware presentation, and perhaps policies that preserve incomparability rather than forcing every candidate set into a total order.

8.6 Domain Grounding Is Both an Inspectable Prior and an Inspectable Error Surface

AIrtist’s domain grounding encodes a recognizable position. Its concept-development policy privileges enacted artistic operation over topical illustration, material and spatial necessity over arbitrary styling, specific spectator relations over generic participation, field differentiation over first-obvious operating formats, and structural repair over additive explanation or spectacle. These commitments enter through the internal quality model, specialized reviews, opportunity representation, and challenger instructions.

The A5-M5 result suggests that making such search priors explicit can be useful. It is preferable experimentally to an opaque instruction such as “make this more creative,” because an explicit prior can be inspected, criticized, edited, and ablated. The system exposes not only a proposed improvement but the terms under which it sought that improvement: what it attempted to preserve, what it treated as the dominant liability, what neighboring format it tried to escape, and what class of repair it refused.

The failure cases show the other side of the same decision. Domain grounding does not remove the error surface; it reshapes it. A search pressure designed to escape familiar tropes can over-transform a direct solution. A demand for material specificity can encourage additional apparatus. A plausible critique can motivate a repair whose added mechanism is weaker than the parent. Brief 038 is particularly instructive: one Structural intervention is locally successful against its exact parent, while two other adopted interventions fail and the complete A5 pool performs substantially worse than M5. The same structured machinery that helps the search on average can therefore push individual lineages past a stronger economical state.

This is important for how domain expertise should be represented in creative agents. A domain-specific prior should not be treated as expert truth added to a generic model. It is better understood as a contestable search position that changes which errors become likely. Making that position explicit creates two advantages: it can improve search under the conditions studied here, and its characteristic failures become available for diagnosis rather than disappearing inside an undifferentiated “creative” prompt.

Contemporary-art installation ideation makes these issues unusually visible because three properties that can remain hidden in more easily scored domains are constitutive here: revision is non-monotonic; relevant qualities are plural and can conflict; and material, spatial, spectator, and institutional relations cannot be reduced to surface correctness. The domain therefore functions not merely as an application area but as a stress test for creative-agent design under weakly specified reward.

The experiments support a correspondingly bounded backend principle: maintain several viable states long enough to compare them, make intervention conditional rather than compulsory, represent the intended transformation explicitly, and audit selection separately from generation. They do not establish how an artist will use those states in practice.

9 Intended Artist-Facing Use: Idea Search as a Substrate for Negotiated Co-Creativity

The preceding discussion established what the autonomous concept-development backend does computationally; this section asks how those properties could become negotiable to an artist. The intended use begins when an artist has a direction but not yet a specification: they may know what matters, which conditions or materials constrain the problem, and which obvious solutions to exclude while remaining uncertain about what the work should become. AIrtist turns one such rough brief into five ranked project hypotheses. The experiment measures that computational substrate; negotiation over its assumptions and outputs remains a subsequent

interaction problem.

9.1 From a Provisional Query to Five Project Hypotheses

A conventional generative interface often treats the prompt as a specification: the user states what is wanted, and the model attempts to produce an answer. This model is poorly matched to early creative work, where part of the difficulty is precisely that the desired answer is not yet known.

As an information-retrieval analogy, we treat the artist's brief as a provisional query into a much larger space of possible projects. The analogy is useful because the brief identifies a local problem without exhausting its solution: its free-text premise states what is at stake; must-address conditions protect pressures the artist does not want development to lose; must-avoid conditions eliminate directions already known to be unwanted; and site, material, institutional, or production requirements further constrain the region to be explored. The resulting input specifies an under-determined development problem rather than a target artifact.

The analogy should not be read literally as document search. AIRTIST does not retrieve an existing artwork or query a database of pre-authored concepts. Its computational mechanism is generative search over proposal states: candidate project directions are constructed, reviewed, transformed, preserved or displaced, and finally ranked. What the artist receives is therefore not a list of retrieved answers but a locally constructed field of possible responses to the brief.

In the evaluated configuration, search after the brief is autonomous and returns five comparatively developed proposal dossiers rather than short brainstorming fragments. Their artistic operation, spatial and material logic, spectator relation, and production implications are explicit as described in Section 4. The dossiers are starting directions for subsequent artistic work rather than five artworks from which the model has identified a correct one; subsequent artistic judgment remains with the artist.

9.2 The Sampled Possibility Set as a Structured Field of Alternatives

Treating the returned proposals as a field of alternatives connects the computational object studied in the experiment to a broader search-and-negotiation account of co-creativity. If a collaborator already knew the exact desired solution, one optimized completion might be sufficient. The more characteristic early-stage case is different: the collaborator needs help making the alternatives themselves concrete enough to think with. The computational role of search is then not simply to maximize the probability of one apparently optimal answer, but to construct a useful field of candidate directions for subsequent human judgment.

RQ1 supports treating this field as a first-class output: the matched-control gain is not confined to one oracle-best candidate (Section 6.1.1). The practical point is not to preserve all five indefinitely, but to avoid treating the surrounding alternatives as disposable scaffolding.

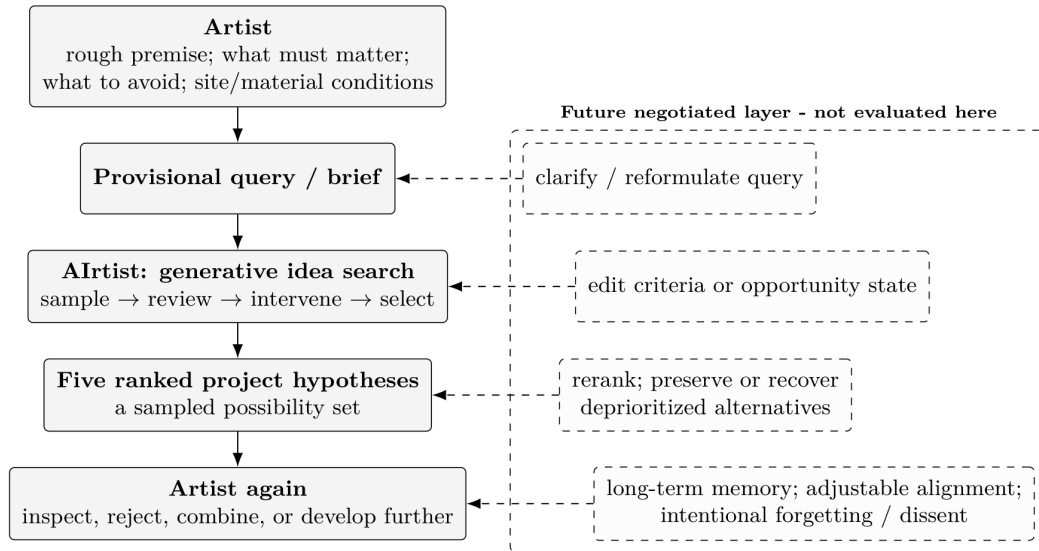


Figure 11: From generative idea search to negotiated co-creativity. In the evaluated system, an artist-authored provisional brief is autonomously expanded into a ranked sampled possibility set of five complete project hypotheses. There is no artist feedback inside the evaluated pipeline. Clarifying questions, editable search states, criteria negotiation, branch recovery, and longitudinal personalization are future interaction layers rather than capabilities evaluated in the present study.

The post-hoc breadth result adds a qualification: several proposals do not automatically constitute broad exploration, and branch or agent count cannot stand in for diversity.

This points toward a richer model of generative idea search. Quality and breadth are potentially separate search objectives. A future system might explicitly diversify candidate operations, cluster related proposals, preserve alternatives from different regions, or maintain a small number of minority directions even when they are not favored by the dominant ranking criterion. The present AIrtist configuration does not implement such a diversity-aware candidate policy; the current result instead motivates testing such policies explicitly.

More generally, a structured field of alternatives need not mean maximum dispersion. Two distant proposals can differ superficially while relying on the same artistic operation, whereas relatively close proposals can expose consequential differences in material causality, institutional procedure, or spectator position. For an artist-facing search system, the relevant problem is therefore not simply to return the most dissimilar five items, but to make several consequentially different directions available without allowing low-quality variation to consume the entire field.

9.3 Ranking as Local Visibility and Attention Allocation

Generating several alternatives creates a second problem familiar from information retrieval: not everything that exists in the candidate collection will receive equal attention.

Ranking is therefore not merely a presentation convenience. Within the sampled possibility set, it determines local visibility: which generated direction is encountered first, which proposal anchors comparison with the others, and which candidates risk being overlooked. At larger cultural and institutional scales, ranking systems can govern visibility in much broader ways. The present experiment makes no such culture-wide claim; it studies the smallest instance of the same structural problem, namely visibility among five candidate project hypotheses produced for one artist-authored brief.

RQ3 supplies the empirical reason to keep ranking provisional: higher-utility measured material often already exists in the generated pool without being surfaced first, while the present Hybrid advantage over Base is unresolved.

An artist-facing interpretation should preserve both findings. Ranking offers one way to manage the non-trivial attention cost of five developed dossiers. But the ordering should remain provisional rather than authoritative. A concise way to state the intended semantics is:

The ranking says which proposal AIrtist recommends reading first, not which proposal the artist should make.

This is also why the intended output remains the ranked set rather than rank 1 alone. Suppressing lower-ranked candidates would give the selector more authority precisely where the

experiment reveals considerable remaining recognition error. Returning all five keeps deprioritized options recoverable and allows the artist to disagree with the system's allocation of attention.

This creates a direct connection between information retrieval and co-creative system design. Relevance is no longer only a property of documents retrieved for an information need; it can govern which generated possibility becomes creatively visible within an interaction. Future systems could expose different orderings of the same candidate field according to different criteria—for example conceptual necessity, novelty, feasibility, public or ethical risk, or distance from a creator's habitual solutions—rather than collapsing these concerns into one hidden score.

The present experiment intentionally does not do this. One bounded internal working model is frozen so that search policies can be compared without simultaneously changing the criterion of preference. Criteria-contestable ranking is therefore an interaction problem for a subsequent system, not an unreported capability of Alrtist.

9.4 Autonomous Search Is Not Yet Negotiated Agency

The evaluated system implements search, but not yet the second half of a broader search-and-negotiation model of human-AI co-creativity.

This boundary is substantive: the current system provides no artist-side query clarification, search-state or criteria editing, branch recovery during a run, or personalization. Section 9.5 develops the corresponding query, search-state, recovery, and longitudinal extensions.

Competitive refinement nevertheless creates some of the computational objects around which such negotiation could later occur. Inside the system, intervention is already contestable in a limited sense: an incumbent remains eligible and a proposed change must earn replacement. Human contestability would add a different level of agency. The collaborator would be able to contest not only the generated candidate but the policy that produced it.

Competitive refinement makes intervention contestable inside the computational search; negotiated co-creativity would make the search itself contestable to the human collaborator.

The opportunity state is particularly useful for negotiation because it externalizes otherwise tacit steering decisions: what the system intends to preserve, change, escape, exploit, and refuse as an adequate repair.

In the present experiment these fields are internal search state. In a negotiated system they could become objects of disagreement. An artist could agree with the diagnosis but reject the proposed escape; preserve a mechanism the model wants to remove; declare that an inefficient material process is essential; or deliberately permit a solution the default quality model treats as too didactic, decorative, spectacular, or familiar.

This would change the role of AI initiative. A proactive system could ask questions, point out contradictions, propose transformations, criticize a direction, or refuse an unsafe implementation. But each of these actions also risks allowing the system to define the artistic problem on the artist's behalf. Negotiated agency therefore requires more than additional conversational turns. It requires making visible when the system is assisting, when it is steering, and what assumptions authorize that steering.

9.5 From Generative Search to Negotiation and Long-Term Collaboration

The present architecture suggests a natural sequence for extending autonomous idea search into negotiated co-creativity.

The first extension is query negotiation. If the initial brief is a provisional query rather than a complete specification, the system need not assume that the first formulation is sufficient. It could detect unresolved constraints, tensions between must-address conditions, or ambiguity about what is intended to remain open, then ask a small number of targeted questions before committing search resources. The important research question is not simply whether clarification improves proposal scores, but when clarification helps articulate an artist's intention and when it instead prematurely determines the problem for them.

The second extension is search-state negotiation. Opportunity states could be exposed before transformation. Artists could edit or lock individual fields, request alternative diagnoses, or ask different agents to formulate competing readings of the same incumbent. A disagreement over what should be preserved could then generate genuinely different searches rather than another rewrite under one hidden interpretation.

The third extension is recoverable selection. Ranking need not permanently erase losing candidates. A future interface could preserve minority alternatives, permit criterion-specific

reranking, display proposals whose pairwise ordering is unstable, and allow the artist to recover a branch that the model deprioritized. The current recognition gap makes this especially important: if selection is fallible, recoverability is not only an interface convenience but a safeguard against premature collapse of the candidate field.

These extensions move AItist toward negotiated agency without requiring that every internal model operation become a conversational turn. Human involvement can be selective. An artist might delegate broad initial search, intervene only when the system identifies a consequential conflict, and reclaim control when a ranking decision would otherwise remove an interesting direction from view.

A further extension is longitudinal. The current AItist has no persistent model of an individual artist. Every user receives the same frozen search machinery and bounded working criteria. Long-term creative collaboration, however, creates an obvious role for memory. Retrieval and personalization could make recurring concerns, materials, rejected clichés, philosophical commitments, past projects, and unresolved ideas available to later searches.

Such memory also creates a distinctive failure mode: aesthetic enclosure. A system that becomes increasingly good at predicting what an artist has previously accepted may progressively reduce the chance of encountering what the artist has not yet learned to want. Personalization can therefore become another form of fixation.

The present results make this a concrete hypothesis for longitudinal testing. A5 already shows that higher measured quality need not coincide with greater latent-semantic breadth. A longitudinally personalized agent should therefore not assume that increasing fit to an artist's prior preferences will preserve productive difference. Future systems may need mechanisms explicitly designed to counter enclosure: different temporal layers of memory, intentional forgetting, adjustable degrees of alignment, or separate agents in which one remains closely aligned while others deliberately retrieve or generate dissenting and unfamiliar directions.

These controls would also have to make creative data governance visible. An artist should be able to inspect what personal material is being treated as precedent, decide what may be retained or forgotten, and distinguish persistent artistic commitments from temporary experiments that should not harden into a permanent profile.

10 Limitations and Threats to Validity

10.1 Construct Validity: Model Preference Is Not Artistic or Curatorial Judgment

The dominant limitation is construct validity. The principal outcomes in this study are derived from blind holistic comparisons performed by GPT-5.6 Sol. Exact order reversal, consensus analysis, verbosity diagnostics, and leave-out tests probe the stability of this measurement procedure. None establishes that model preferences coincide with those of professional artists, curators, producers, institutions, or publics.

This boundary is particularly important because two model-mediated evaluative structures appear in the system and should not be conflated. AItist's internal ten-dimensional working model shapes review and ranking through explicit concerns such as prompt fidelity, enacted tension, artistic-operation necessity, field distinctiveness, spatial and visual encounter, material necessity, spectator specificity, reception resilience, and production/institutional viability. The external evaluator, by contrast, performs a separate holistic A/B judgment of complete proposals. It is instructed to compare the artistic proposition as a whole rather than mechanically reproduce the internal scorecard. The two instruments overlap in the concerns they attend to, but they are not the same measurement.

Consequently, the main results establish comparative preference under a disclosed model-based evaluation instrument; they do not validate the internal quality model as a general theory of contemporary art. Nor does separating the generator from the external evaluator solve the larger construct-validity problem. The separation is important experimentally because external utility is unavailable to the generator, lane selector, and terminal rankers, but the resulting preferences remain model-mediated rather than professionally validated artistic judgments.

The internal working model itself is intentionally normative. It favors operational necessity, material and spatial specificity, explicit spectator relations, field differentiation, resistance to generic additive spectacle, reception resilience, and plausible institutional realization. These are defensible pressures for the constrained installation-ideation problem studied here, but they are neither neutral nor exhaustive. Practices organized around decorative excess, explicit pedagogy, social service, narrative directness, spectacular display, devotional function, deliberate impracticality, or institutional antagonism may be valued differently by other artistic and

curatorial positions. Making these commitments explicit improves inspectability; it does not turn them into universal criteria of artistic quality.

The same limitation applies to model-generated reasons. Judge rationales are useful in the Mechanism and Failure Analysis because they reveal what the evaluator says it attended to—for example, a material violation, excessive apparatus, or an unclear spectator relation. They are not an independent source of expert interpretation and should not be treated as causal explanations of why one proposal is artistically superior. Verdict and explanation originate from the same model-mediated instrument.

The strongest claim the study therefore supports is intentionally narrower than “AIrtist makes better art”: under a frozen experimental setting and external holistic model evaluator, its candidate sets and selected lineage interventions exhibit the measured differences reported in Section 6. Establishing correspondence with professional artistic judgment requires a separate human validation study.

10.2 External and Ecological Validity: A Curated Benchmark of Unrealized Proposals

The study evaluates 50 held-out commission-style briefs, IDs 026–075, disjoint from the 25 briefs available during system development. The main matched-control direction survives removal of every individual brief and every complete coverage category. This supports the aggregate comparison on the held-out benchmark, but it should not be confused with evidence from a natural sample of contemporary-art commissions.

The briefs were constructed within the project to cover five deliberately balanced forms of pressure: thematic/conceptual, site-specific, material-constrained, institutional/production-constrained, and anti-cliche/difficult. These categories are useful for coverage and sensitivity analysis, but they are not an empirical taxonomy of contemporary-art practice, and their equal frequency is artificial. The experiment can therefore support aggregate claims about performance on this controlled set of constrained installation problems. It cannot estimate how often the same advantage would occur across naturally distributed museum commissions, artist-authored project notes, open calls, gallery briefs, or other cultural contexts.

The benchmark size also places a specific boundary on subgroup interpretation. Each coverage category contains ten briefs. Leave-one-category-out analysis is useful as a robustness check of the overall matched-control direction; it is not evidence that AIrtist performs differently across artistic categories. Likewise, individual success and failure cases should not be interpreted as prevalence estimates for particular artistic mechanisms.

More importantly, the evaluated objects are textual project hypotheses, not realized installations. A dossier can make a coherent case for material behavior, scale, circulation, duration, maintenance, public encounter, staffing, and budget and still fail when produced. No engineering study, conservation review, accessibility review, fabrication process, installation period, maintenance cycle, commissioning decision, or audience observation validates the proposed works.

This text-to-work gap is unusually consequential in installation practice. Sound behaves differently in an actual room; materials wear unpredictably; visitors ignore intended routes; access and fire requirements alter spatial arrangements; natural light changes; repeated interaction modifies the work; and institutional labor that appears minor in prose may become constitutive in practice. Textual proposal evaluation can reason about such relations only counterfactually.

The post-hoc renderings do not close this gap. Their purpose is to make transformations in source dossiers visually legible—for example, a boundary moving from a frontal condition into a walkable field. Image generation can also supply convincing architecture, fabrication detail, lighting, scale, and finish that the text did not earn. Renderings are therefore explanatory visual hypotheses, were never shown to the evaluator, and provide no independent evidence that the represented work would function as depicted.

10.3 Internal Validity and Mechanism Attribution

M5 substantially matches the iterative scaffold but is not compute-identical, and several domain-structured components change together. The A5–M5 result is therefore a package-level comparison; component attribution requires targeted ablations.

RQ2 likewise supports only a conditional claim: adopted replacements are preferred on average to their exact parents, but no matched forced-replacement policy was generated. The experiment therefore cannot show that allowing retention itself causes higher quality or that competitive refinement is superior to mandatory replacement.

RQ3 cleanly isolates ranking because Base and Hybrid receive identical candidate pools, but the observed Hybrid-minus-Base Utility@1 advantage is not statistically resolved. The supported claim is therefore that selection is a separable recognition problem, not that the present tournament is a reliably superior solution.

Finally, all core generation conditions use the same base generation model family. This is an advantage for the within-study comparison because variation in base-model capability does not confound the search-policy contrasts. It leaves model-generalization open. A stronger generator might reduce the marginal value of structured refinement; a weaker generator might benefit more from it; and a differently trained model might respond differently to field-escape or preservation instructions. Cross-model generation experiments are required before treating the architecture as model-independent.

10.4 Evaluation-Procedure Dependence and Exploratory Analyses

The primary external evaluation is exhaustive within each brief but not composed of independent judgments at the pair level. Each logical A5–control pair is presented in exact A/B-reversed orientations, but multiple comparisons within one pass are processed inside the same fresh Temporary Chat. Opposite orientations are isolated in separate chats, making order instability observable, but comparisons within a pass can still be affected by contextual carryover.

For this reason, the study treats the brief rather than the candidate pair as the inferential unit. The 7,500 physical judgments provide dense measurement of 50 held-out tasks; they do not constitute 7,500 independent benchmark observations. A stricter future design could place every logical comparison—or small randomized micro-batches—in an independent context.

Exact swaps expose rather than eliminate evaluator instability. Overall swap agreement is 79.41%, rising to 94.73% among pairs where both orientations report confidence 4–5. Consensus-only analysis shows that the matched-control direction does not depend on the comparisons whose verdict visibly reverses under presentation order, but selecting only stable pairs changes the estimand and may preferentially retain easier distinctions. It should therefore remain a sensitivity analysis rather than be interpreted as a cleaner ground truth.

Relative-length analysis addresses another identifiable threat but should not be overread. A5 dossiers are somewhat longer than M5 dossiers, while the association between relative length and preference is weak and the A5 direction persists when A5 is not longer or is strictly shorter. This makes a simple monotonic verbosity explanation difficult to sustain. It does not establish that the holistic model judge is free of subtler stylistic biases involving fluency, curatorial vocabulary, structural completeness, tone, or apparent professional confidence.

Several analyses in the paper are explicitly post-hoc. Latent-semantic pool breadth, the relationship between quality effect and breadth difference, parent-to-Focused versus parent-to-Structural semantic distance, and the pointwise-to-tournament stage diagnostic were introduced to understand the frozen experiment rather than as members of the planned primary endpoint family. Their role is mechanistic and hypothesis-generating. In particular, lower A5 breadth under one semantic representation should not be generalized to diversity in every artistic sense, and the observed quality–breadth relation does not establish that concentration causes the measured quality gain.

The qualitative cases are selected for explanatory contrast rather than prevalence estimation. They illuminate mechanisms measured elsewhere and add no independent statistical evidence.

Alrtist’s field-context construction introduces a further information-quality boundary. The system can use model-mediated recall of artists, works, texts, and neighboring formats to construct a shared field map. This is associative context construction rather than provenance-bearing retrieval. Such recall can be incomplete, inaccurate, culturally narrow, or anachronistic. Recalled precedents are used as internal prompts for differentiation rather than as factual art-historical evidence in the external evaluation. A stronger future system should separate associative recall from verifiable retrieval and expose provenance when factual claims about artistic precedent matter.

10.5 Human Co-Creative Validity Is Outside the Present Experiment

The intended artist-facing use model makes a final boundary especially important. The present study evaluates an autonomous concept-development backend, not negotiated human–AI co-creativity.

The experiment does not observe whether artists find the returned project hypotheses generative, whether they prefer the structured set to alternatives produced by another system,

whether they combine or reject proposals, whether rank position causes fixation, or whether the opportunity representation helps them articulate their own intentions.

Accordingly, improved sampled-set quality should not be rewritten as improved human creativity, creative satisfaction, authorship, trust, agency, or co-creative experience. Likewise, returning several proposals creates the possibility of comparison but does not establish that five is the right number for a working artist, that ranking reduces cognitive load, or that an artist would benefit from seeing all five rather than a differently organized field.

Future validation should therefore extend in at least three directions. First, professional artists and curators should compare proposal sets and individual interventions to test construct validity of the model-mediated preferences. Second, longitudinal interaction studies should test whether multiple ranked alternatives, editable search states, and recoverable branches actually support exploration rather than fixation or cognitive overload. Third, selected proposals should be followed through technical review and, eventually, fabrication and situated encounter. These studies answer questions that no larger model-judge benchmark can substitute for.

11 Conclusion

A consequential part of contemporary-art practice occurs before an artwork has stabilized: an artist may know what problem, pressure, site, material condition, or institutional constraint matters without yet knowing what the work itself should become. This paper has treated that interval as an explicit computational task. AIRTIST develops an under-specified artist-authored installation brief into several complete project hypotheses and models development as contestable transformation: proposed changes must compete with the artistic state they are intended to replace rather than being accepted simply because additional inference has produced them.

The empirical results support this formulation, but not uniformly across all stages. AIRTIST produces a stronger measured five-proposal set than all three planned controls, including the substantially matched M5 condition that starts from the same five self-refined parents and closely matches the iterative scaffold. Against M5, the centered complete-pool effect is $+0.0836$ (95% CI $[+0.0340, +0.1304]$, Holm $p = .0015$). The gain is therefore difficult to explain as a consequence of better starting samples or merely generating more generic continuations from the same parents. At the intervention level, 59.6% of final lane states remain content-identical to their incumbents; among the 101 replacements actually adopted, challengers receive mean external preference 0.693 against the exact states they displace, with a brief-balanced centered effect of $+0.437$. These results do not identify which individual reviewer, opportunity-state field, or intervention role is causally necessary, and they do not establish that incumbent retention is superior to forced replacement. They do show that domain-structured development can improve the proposal field while preserving non-intervention as a legitimate outcome.

Terminal selection is less resolved. The Hybrid selector has favorable point estimates relative to Base, but its $+0.0533$ Utility@1 advantage is not statistically resolved. Stronger candidates often already exist inside the generated five-proposal pool without being surfaced first. The experiment therefore establishes the recognition problem more clearly than it establishes the present solution: producing promising material and recognizing which existing candidate deserves attention are empirically separable capabilities.

This separation is the broader methodological lesson of the study. A weak surfaced proposal can arise because the system failed to construct strong alternatives, because an intervention damaged a stronger state, or because ranking hid useful material that had already been generated. Evaluating only one final answer collapses these failure modes. AIRTIST instead makes candidate-set quality, intervention quality, and attention allocation separately measurable. The post-hoc breadth analysis adds a related caution: the higher measured quality of A5 relative to M5 did not coincide with greater measured breadth under the chosen latent-semantic representation. Better proposal sets and broader exploration should therefore be treated as distinct objectives rather than assuming that more agents, branches, or outputs automatically create a wider creative field.

Contemporary-art installation is central to this argument rather than merely a difficult benchmark. In this domain, a change in material causality, spatial organization, spectator position, or institutional procedure can alter the identity of a proposal rather than simply improve its presentation. A critique can correctly diagnose a weakness while the proposed repair destroys the strongest relation in the parent. This is why preservation, transformation, and selection need to remain visible as separate decisions. Similar problems arise in other open-ended creative domains, but installation practice makes their consequences unusually explicit.

The claims of the present study remain bounded. The evaluated objects are textual project hypotheses rather than realized artworks; the principal quality signal is model-mediated; and no artist negotiates with the system inside the frozen experimental pipeline. The results therefore concern a computational development policy under a disclosed working quality model, not professional curatorial consensus, cultural value, or demonstrated improvement to human creativity. The next step is to expose the same development process to negotiated co-creativity, so that artists can contest diagnoses, preservation constraints, candidate priorities, and discarded branches rather than receiving only a finished ranking.

The aim is not to automate the decision of what the artwork should be. It is to make the space in which that decision is formed richer, more inspectable, and less prematurely collapsed.

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